

Real-Time Prosodic Tennis Commentary via Streaming Vision-Language Models

by

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ABSTRACT

Major tennis tournaments feature hundreds of matches across multiple courts, yet human commentators can only cover a fraction of them. This leaves fans without play-by-play coverage for most matches, particularly those on side courts. This thesis presents a system for generating real-time, emotionally expressive AI commentary for tennis matches, addressing both the content generation and prosodic delivery challenges that distinguish engaging sports commentary from robotic narration. We develop two complementary approaches. First, we create a modular pipeline that processes tennis clips through a computer vision module to extract shot metadata, uses a large language model to generate commentary, and synthesizes speech using a prosodic text-to-speech model. This pipeline incorporates an excitement modulation mechanism to control both the linguistic style of the generated text and the prosodic characteristics of the synthesized speech. Second, we work toward a unified approach by fine-tuning StreamingVLM, a vision-language model trained on over 4,000 hours of sports commentary but lacking tennis data. To bridge this gap, we combine commentary from the modular pipeline with a novel scalable pipeline that extracts and cleans commentary from YouTube streams via optical character recognition, automatic speech recognition, and large language model based cleaning. Initial fine-tuning experiments demonstrate that StreamingVLM can learn to generate tennis-specific commentary with correct player identification, appropriate shot descriptions, and accurate score tracking. While limitations remain in commentary cadence, factual accuracy and long-term coherence, these results establish a foundation for further development toward high-quality, emotionally resonant automated tennis commentary.

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Chapter 1

Introduction

1.1 Motivation

The number of matches played during major tennis tournaments makes it almost impossible for human commentators to cover every court in real time. At tournaments like the US Open and Wimbledon, matches on side courts often receive no live commentary or broadcast, leaving fans uninformed unless they attend in person or follow text-only updates. There is a need for automated solutions to narrate events in such matches, providing play-by-play commentary to make engagement more accessible and enhance fan experience.

1.1.1 Industry Context: AI in Tennis

IBM is transforming how tennis fans and players interact with the game. Through longstanding partnerships with the United States Tennis Association (USTA) and the All England Lawn Tennis Club (Wimbledon), IBM delivers AI-powered experiences to millions of fans around the world [1,2]. These partnerships have yielded a suite of AI-powered innovations. Match Chat is an interactive assistant that can answer fans' questions live during singles matches. SlamTracker offers dynamic win probability projections that refresh throughout each match based on momentum, expert analysis, and player statistics. Key Points generates concise three-bullet summaries of articles, tournament data and match analysis for fans seeking quick updates. Figure 1.1 illustrates these features as deployed.

Beyond fan engagement at professional tournaments, IBM is partnering with Agassi Sports Entertainment (ASE) to democratize access to racquet sports for players at all levels. In November 2025, ASE and IBM announced a multi-year partnership aimed at transforming the global racquet sports landscape through next-generation AI and digital innovation [3]. Branded as "Agassi Intelligence", the collaboration leverages IBM's watsonx.ai platform and

computer vision technology to build tools to analyze athletic movement from everyday footage, enabling professional-grade coaching to be delivered directly to players’ mobile devices. Andre Agassi, co-founder of ASE and eight-time Grand Slam Champion, described the initiatives mission as delivering “smarter tools, stronger communities and global access to the sports we love” [3].

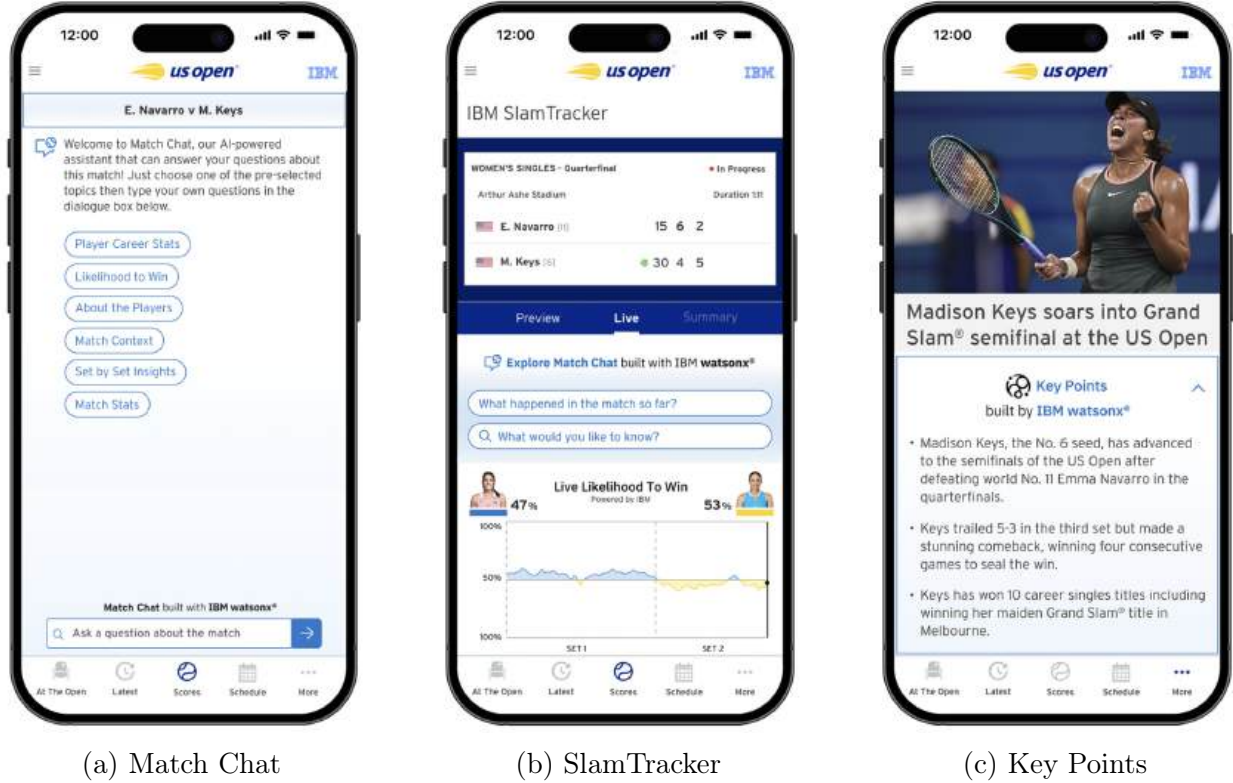


Figure 1.1: IBM’s AI-powered fan engagement features at the US Open [1]. (a) Match Chat: an interactive assistant that answers fan questions live during matches. (b) SlamTracker: dynamic win probability projections that refresh throughout each match. (c) Key Points: concise three-bullet summaries of match analysis and tournament data.

1.1.2 Fan Demand for Enhanced Real-Time Viewing Experiences

There is strong fan demand for richer live viewing experiences, including real-time updates and commentary. A June 2025 global survey conducted by Morning Consult and commissioned by IBM surveyed over 20,000 sports fans across 12 countries, encompassing a wide variety of sports including tennis. The findings reveal that 86% of tennis fans worldwide value AI-powered features, especially those providing real-time analysis and personalized highlight content [4].

The survey highlights a clear preference for live content consumption. Across all sports, fans are most likely to tune-in live from home for their favorite sporting events, and live streaming apps rank among the most popular choices for the 73% of fans who use mobile sports apps [4]. When asked about AI-driven enhancements, real-time updates emerged as the most desired improvement to sports engagement, with 35% of fans rating them as a high priority [4]. For fans attending matches in-person, real-time commentary also remains a top priority, underscoring the importance of timely, informative narration regardless of viewing context [4].

While previous iterations of IBM’s analytics offerings have provided post-match AI Commentary for highlight videos [1], the survey findings suggest an opportunity to extend AI-driven commentary into the live viewing experience. This demand motivates our work on developing a system capable of generating real-time, accurate and emotionally expressive commentary that aligns with the visual events of a match as it unfolds.

1.1.3 From Content Generation to Expressive Delivery

Purohit (2023) introduced an AI commentator that processes tennis match videos to produce real-time, shot-by-shot commentary [5]. This system combines a computer vision module for fine-grained rally understanding with a data-to-text language model to narrate what is happening on court. The AI commentator was shown to generate informative and fluent descriptions of tennis clips and was even slated for trial implementation at the Wimbledon Championships [5].

While this prior system successfully narrates the visual events of a rally, its focus was on visual perception and content generation rather than the delivery of the commentary. Human commentators do not merely describe actions; they also modulate their timing, pace, and tone to match the ebb and flow of the game. For example, crucial points are called with heightened excitement, and the cadence of speech aligns with the tempo of play. These aspects of prosody and emotional expression were beyond the scope of the initial commentary system. The generated commentary in Purohit’s work, delivered as text and later synthesized to speech, was not explicitly aligned in real-time to the video, nor did it vary its emotion with specific insights from the play. Thus, a gap remains in making AI-generated commentary both precisely aligned to on-court action and as natural and engaging as human-delivered commentary.

1.2 Contributions

This thesis presents a system for generating real-time, emotionally expressive AI commentary for tennis matches. To the best of our knowledge, this work is the first to introduce an excitement-driven AI commentary framework that specifically models and controls prosodic expressiveness as a function of point-level match dynamics. Building on prior work in visual perception and content generation, we address the gap in prosodic delivery and extend the system to leverage state-of-the-art vision-language models. Figure 1.2 presents the two main components of this thesis, contrasting the previous modular approach with the proposed unified approach.

First, we develop a pipeline for excitement-driven prosodic commentary, shown in the middle column of Figure 1.2. This pipeline processes isolated tennis clips through a computer vision module to extract shot metadata, then uses a large language model (LLM) with post-processing to generate timestamped commentary. The system classifies points into three excitement levels based on rally characteristics and maps these to fine-grained prosody tags for each sentence, enabling text-to-speech (TTS) synthesis that modulates tone and pacing based on match intensity. This approach was demonstrated at the US Open Behind the Scenes session, where it showcased the shift from monotone AI commentary to expressive, human-like delivery. This process is described in Chapter 3.

Second, we create a dataset for fine-tuning StreamingVLM, a large vision-language model trained on over 4,000 hours of sports commentary but lacking tennis data. We utilize and develop two complementary data generation methods: (1) a computer vision and LLM pipeline, based on the methods described in Chapter 3, that processes isolated tennis clips with directed video-to-commentary alignment, and (2) a scalable four-stage pipeline that extracts and cleans commentary from YouTube streams of tennis matches. The latter approach segments matches at point boundaries detected via optical character recognition, aligning with StreamingVLM’s chunk-based training strategy while ensuring semantically coherent segments. Additionally, we utilize automatic speech recognition and LLM-based cleaning methods to extract and refine commentary text for each segment. This process is described in Chapter 4.

We demonstrate initial progress toward a unified system that combines visual understanding with expressive commentary generation, shown in the right column of Figure 1.2. By fine-tuning StreamingVLM on our tennis dataset, we aim to produce a model capable of generating contextually aware, real-time commentary directly from video input, collapsing the separate vision and language stages. This unified approach has the potential for integration with our prosodic TTS pipeline to deliver natural, emotionally engaging audio commentary.

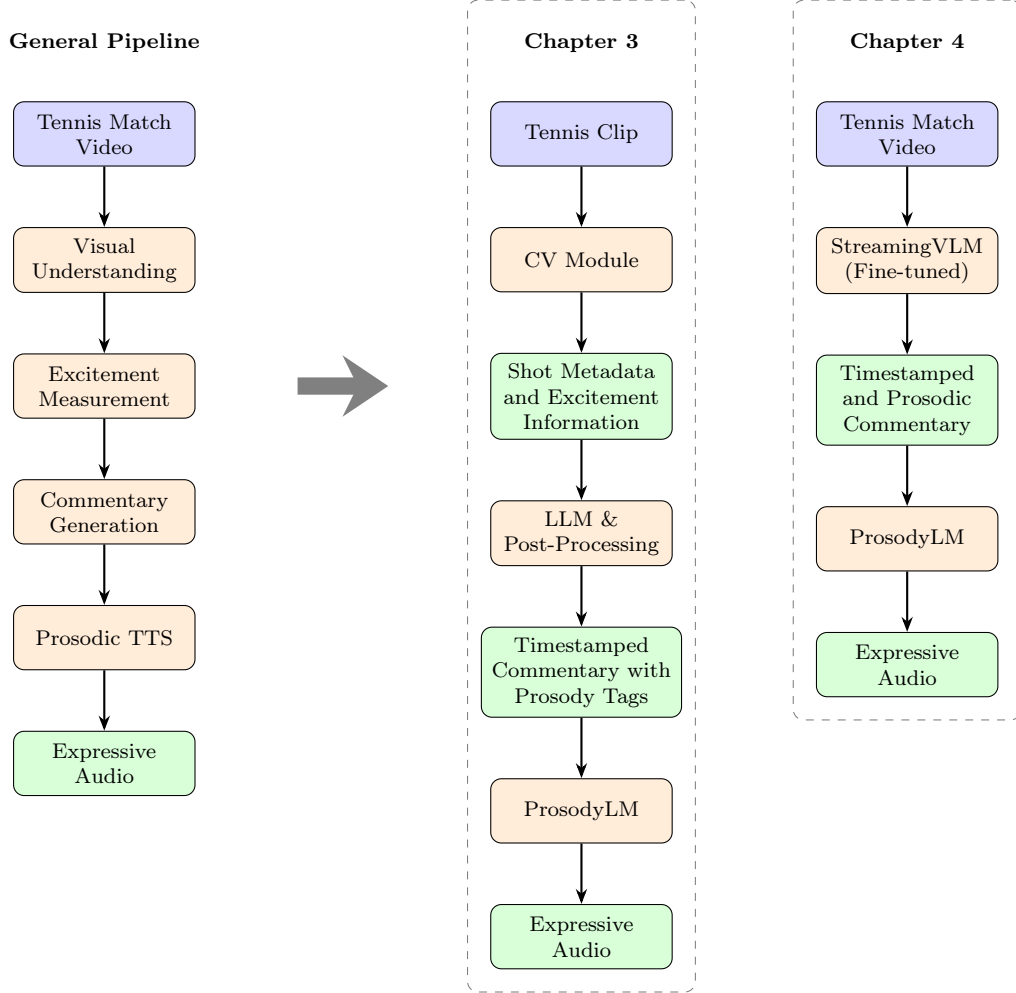


Figure 1.2: System architecture overview. **Left:** The core pipeline abstraction for automated sports commentary, showing the flow from video input through visual understanding, excitement measurement, commentary generation, and prosodic TTS to produce expressive audio. **Center (Chapter 3):** The modular pipeline from Chapter 3, which processes isolated tennis clips through separate vision and language stages, followed by prosodic TTS via ProsodyLM. **Right (Chapter 4):** The unified approach from Chapter 4, where a fine-tuned StreamingVLM combines visual understanding and commentary generation into a single model.

Chapter 2

Background

A recent analysis of 1,209 publications found that the field of AI in sports has experienced explosive growth since 2015, with five central research themes emerging: physical performance assessment, skill optimization, health monitoring and injury prevention, tactical analysis and referee assistance, and immersive event experiences [6]. Estimates value the global AI in sports market at USD 7.63 billion in 2025, and it is projected to reach USD 26.94 billion by 2030, reflecting a 28.69% compound annual growth rate [7]. This growth is driven by enhanced investment in edge computing, optical-tracking systems and generative content engines, with computer vision latency and accuracy improvements propelling automated event detection [7].

Within this landscape, automated sports commentary presents a unique challenge. It requires not only robust visual understanding but also the generation of natural, emotionally appropriate speech synthesized to on-court action. This thesis draws on advances in computer vision, natural language generation, and speech synthesis, identifying gaps in aligning visual events with the emotional prosody of commentary.

2.1 Visual Understanding of Tennis Matches

Robust visual understanding of sports is a foundation for automated commentary. The integration of computer vision with sports has transformed competitive, educational and recreational practices, propelling them into an intelligent era [6]. Across professional leagues, computer vision and sensor analytics are used to capture player movements, ball trajectories, and key events [8] [9] [10] [11]. In tennis, vision systems typically perform tasks such as court localization, player and ball tracking, pose estimation, and action recognition. For example, Purohit’s AI commentator system employed a single-camera vision module that could detect the court and net boundaries, track both players and the ball, estimate player poses, and

classify shots [5]. Similarly, SwingVision is an iOS application that leverages computer vision techniques to analyze tennis matches from smartphone video in real-time. Recent validation studies have shown that SwingVision produces highly accurate data when recorded from optimal angles, demonstrating its potential as a lean on-device tool for enhancing visual understanding of tennis matches [12]. Working in a similar space, SportAI is a Norwegian startup founded in 2023 that uses computer vision and machine learning to generate biomechanical analyses from standard smartphone video to build detailed 3D visuals of tennis playing style [13,14]. The system measures things like swing curve, power generation and ball contact point to map the kinetic chain when making a shot, insights previously accessible only to elite players with dedicated performance analysts [14].

The deployment of computer vision in tennis continues to expand. In 2025, Wimbledon joined the US Open and Australian Open in adopting fully automated electronic line-calling for all matches, replacing 300 human line judges for the first time in the championships 147-year history [13,15]. Hawk-Eye’s ball-tracking system, which uses about 12 cameras to track the ball’s trajectory at 340 frames per second with a margin of error of just 3.6mm, now determines all in/out calls without human intervention [15,16]. In late 2025, the Indian sports-tech startup KPro.ai was named the AI partner for the Aryna Sabalenka vs. Nick Kyrgios exhibition match in Dubai, deploying smartphone-based computer vision to analyze player movement, reaction time, and execution in real-time without wearables [17].

In the domain of soccer, FootyVision represents a recent advancement in comprehensive visual understanding. Built on a YOLOv7 backbone, FootyVision integrates player and ball detection, multi-object tracking, and top-down localization through perspective transformations derived from activation maps [18]. Its architecture enables robust tracking of players and the ball, even under challenging broadcast conditions. Although originally developed for soccer, a similar architecture could be highly beneficial for tennis, where dynamic court perspectives, occlusions, and fast-paced player movements pose major challenges. For generating commentary, the better the system recognizes what happens in each rally, the more accurate and detailed the commentary can be.

2.2 Automated Commentary Generation

Early attempts to automatically generate sports commentary relied on templated narratives driven by live data. For example, Kumano et al. (2019) presented a system that takes structured sports data (e.g. play-by-play logs) and populates predefined commentary templates, outputting synthesized speech descriptions of the game situation [19]. They demonstrated that template-based commentary can effectively convey game state and help listeners follow

the action [19]. While reliable, such systems lack variety and may sound robotic. Recent research has therefore turned to more flexible natural language generation, including LLMs, to produce commentary that is fluent and varied.

In her thesis, Purohit leverages computer vision to extract detailed shot-level metadata, and integrates this with match context from Wimbledon to form a structured prompt for a language model [5]. The language model was finetuned on synthetically generated commentaries to produce realistic outputs [5]. A similar line of work explores customizing open-source LLMs for the sports domain. Cook and Karakuş (2024) propose LLM-Commentator, fine-tuning compact transformer models on soccer event data to generate match commentary [20]. Their results show that when properly fine-tuned on domain-specific datasets, LLMs can produce coherent play-by-play narration from raw match data.

Another key challenge is aligning commentary text with the visual timeline of the game. Rao et al. (2024) highlight that many video-caption datasets suffer from misalignment between action and commentary [21]. To address this, they curated a new dataset, MatchTime, with corrected timestamps for soccer videos and trained a commentary model on it, dubbed MatchVoice. The aligned training data led to substantial improvements and their model achieved state-of-the-art quality, confirming that better temporal alignment of commentary with events significantly boosts generation performance [21].

Overall, automated commentary generation has made rapid progress, from rule-based systems to LLM-driven narratives. Yet, most efforts have focused on what is said rather than how it is said. The delivery of commentary, especially the emotional tone and prosody that convey excitement or tension, remains relatively under-explored.

2.3 Vision-Language Models for Unified Visual Understanding and Commentary Generation

Vision-language models (VLMs) are poised to unify the visual understanding and text generation components of automated commentary into a single end-to-end system. VLMs combine visual encoders with language models to jointly process images or video and generate text, enabling systems that can both perceive and describe sports action. As Li et al. (2025) note in their survey of state-of-the-art VLMs, older architectural approaches such as CLIP primarily trained vision and text encoders jointly from scratch, learning to align images and text in a shared embedding space [22]. More recent methods such as LLaVA and Qwen-VL instead leverage pre-trained LLMs as a backbone, connecting vision encoders to the language model’s existing knowledge and reasoning capabilities [22].

Multi-Modal Large Language Models (MLLMs) extend upon VLMs by incorporating additional modalities such as audio and structured metadata alongside video. Recent work has applied MLLMs to sports commentary. SoccerComment integrates MLLMs with retrieval-augmented generation and a multi-modal clustering memory unit to enhance inference efficiency [23]. TimeSoccer introduces an end-to-end MLLM for soccer that jointly predicts event timestamps and generates captions, using motion-aware frame compression to process full 45-minute match halves [24]. Our work aims to achieve similar commentary generation capabilities utilizing just a VLM by incorporating aspects of structured metadata into the training commentary.

A key challenge for applying VLMs to live sports commentary is processing continuous streams without escalating latency and memory usage. Standard full-attention mechanisms incur quadratic computational costs and degrade on long videos, while naive sliding window approaches either break temporal coherence or suffer from redundant recomputation due to lack of overlap. StreamingVLM, proposed by researchers in the MIT Han Lab in late 2025, addresses this challenge by maintaining a compact KV cache that reuses attention sink states alongside a short window of recent vision tokens and a longer window of recent text tokens [25]. Built on Qwen-2.5-VL-7B Instruct, their model is trained with full attention on short, overlapping video chunks. Then at inference time, it employs the attention sink and sliding window strategy to handle effectively infinite video streams. StreamingVLM was trained on a sports commentary dataset of over 4,000 hours covering basketball, soccer, ice hockey, baseball and American football, but notably not tennis [25]. Additionally, while StreamingVLM demonstrates strong capabilities in generating real-time commentary text, it does not address prosodic delivery. The model achieves real-time processing at 8 frames per second on a single H100 GPU and a 66.18% win rate against GPT 4o mini on a benchmark of videos averaging over two hours in length. As described in Chapter 4, we build on StreamingVLM and adapt it to the tennis domain.

2.4 Prosody and Expressive Speech Synthesis

A distinctive feature of live sports commentary is its expressive prosody – the rise and fall in excitement, the dramatic pauses, and the emphatic tone that matches the game’s intensity. In the context of sports commentary, this means an AI system would ideally raise its voice intensity for dramatic moments or speak calmly during routine play. Achieving this level of expressiveness in synthesized speech is challenging. Prior work has approached the problem by explicitly annotating and modeling prosody in commentary corpora. Brognaux et al. (2013) developed a two-tier prosody annotation scheme tailored to sports commentary, marking

both macro-prosody (global intonation patterns, e.g. excitement vs. calm narration) and micro-prosody (local pitch and stress patterns) [26]. These annotations were used to train an HMM-based speech synthesizer that could generate commentary with varying intonation appropriate to the game context, which led to more natural-sounding commentary compared to a flat, neutral delivery [26]. This early work, however, required laborious manual labeling of speech and was limited by the synthesis technology of the time.

Modern neural text-to-speech (TTS) systems offer new ways to model and control prosody. Neural TTS models such as Tacotron and FastSpeech allow adjustments to speech rate, pitch, and emphasis [27] [28]. Researchers have introduced style tokens and other mechanisms to have a single TTS voice adopt different emotional tones (e.g. excited, sad, formal) [29] [30]. Skerry-Ryan et al. (2018) introduced an extension to the Tacotron neural TTS that learns a latent embedding for prosody from reference audio clips [31]. In their framework, one can supply an example utterance (such as an excited exclamation) as a style reference, and the TTS model will transfer that intonation to a new commentary line [31]. More recently, ProsodyFlow advances this direction by integrating large self-supervised speech models with conditional flow matching, using a speech LLM to extract acoustic features that are mapped into a prosody latent space before generating text-conditioned prosodic vectors [32]. These advances in neural TTS and prosody modeling are promising for expressive commentary synthesis.

A remaining challenge for live sports is determining the appropriate prosodic style on the fly. Recent work by Yang Zhang and Kaizhi Qian, who I am collaborating with on the work presented in this thesis, introduces ProsodyLM. This speech-LM learns emergent prosody control through pre-training by interleaving text tokens with word-level prosody tokens [33]. ProsodyLM shows strong zero-shot abilities to (1) condition its intonation on prompts (e.g. “read this line in an excited tone”), (2) infer speaker emotion and emphasis from prosodic input, and (3) maintain style consistency across turns. These emergent capabilities point toward a future where prosody-aware language models could learn to generate contextually appropriate prosodic variation directly from training data, potentially reducing reliance on on handcrafted labels or post-hoc style transfer. In this thesis work, ProsodyLM, with some modifications, is used to produce the final spoken commentary.

Chapter 3

Developing Prosodic and Timestamped Commentary

In the initial phases of the project, we focused on extending the system described in Purohit’s thesis [5]. The system consists of a computer vision module, a commentary module, and a text-to-speech (TTS) module. Our main focus was improving the commentary and TTS modules in order to produce accurate, expressive and temporally-aligned spoken commentary. As shown in Figure 3.1, our revised pipeline takes point clips and tournament metadata as input, processes them through the computer vision module from Purohit’s thesis to extract shot classifications and directions, and combines this information with tournament metadata. The combined metadata is then passed to an LLM for commentary generation, followed by post-processing to add precise timestamps and prosody tags. Finally, we use ProsodyLM, a TTS model with enhanced prosody processing capabilities, to generate the spoken commentary [33]. We also integrated an excitement control mechanism that links the games context to the commentary tone in a systematic way. The following sections describe each stage of this pipeline in detail.

3.1 Metadata Pre-Processing

3.1.1 Computer Vision Output

As mentioned, we utilize the computer vision module from Purohit’s thesis [5]. This module takes in an input rally video and outputs granular shot-specific data in a raw format that must be transformed before it can be used for commentary generation. Figure 3.2 shows an example of the components of this raw output that we are interested in.

The shot candidate ranges are a list of frame intervals for each detected shot, where each

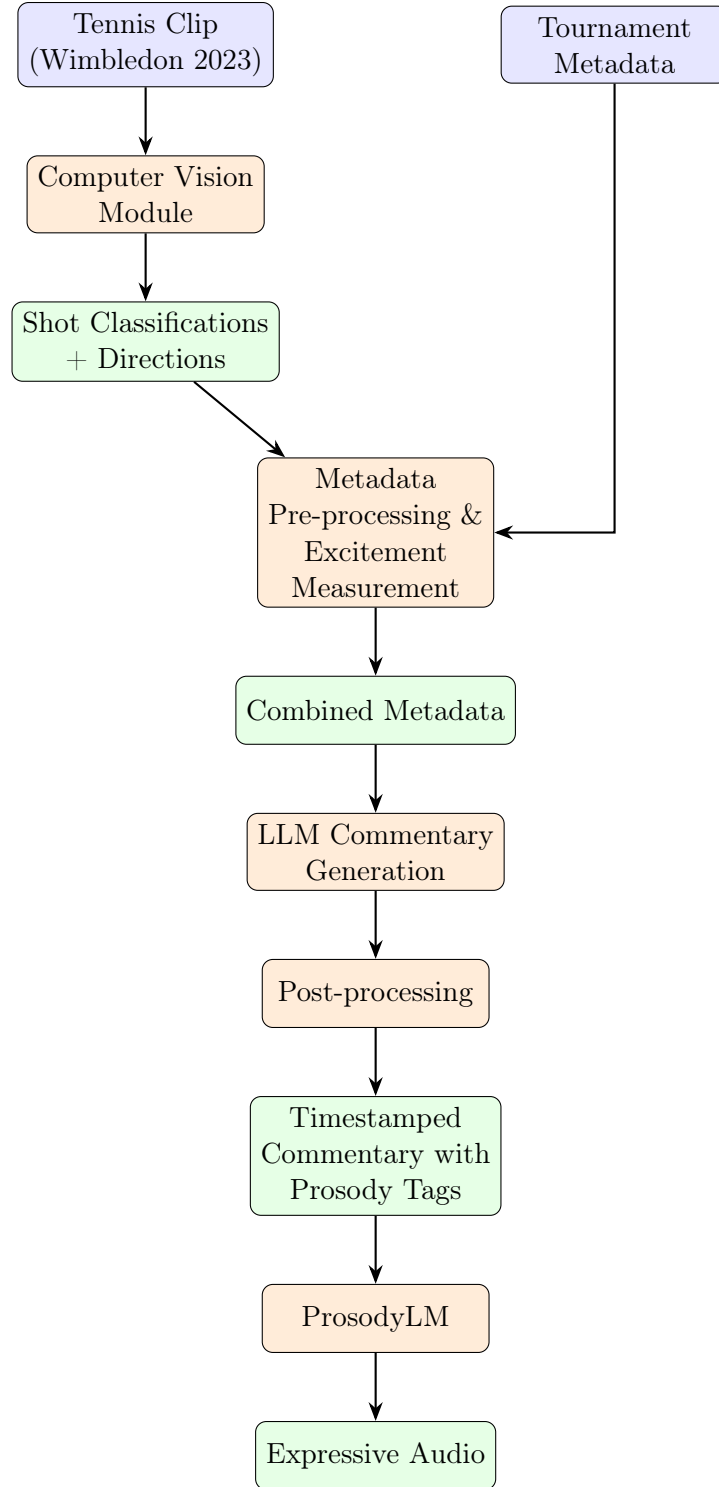


Figure 3.1: Pipeline for generating spoken commentary. Match clips and tournament metadata are processed through the computer vision module, combined into structured metadata, passed to an LLM for commentary generation, post-processed to add timestamps and prosody tags, and finally synthesized into spoken commentary using ProsodyLM.

Raw Computer Vision Output

```
shotCandidateRanges: [[100, 124, "bottom"], [134, 153, "top"], [162, 178, "bottom"],  
                      [189, 211, "top"], [216, 239, "bottom"], [251, 285, "top"],  
                      [342, 374, "bottom"]]  
  
fps: 25.0  
total_frames: 425  
annotationStartFrame: 0  
shotClassifications: ["Serve", "Forehand", "Backhand", "Backhand", "Backhand",  
                      "Backhand", "NotAShot previously in net"]  
shotDirections: ["", "up the middle", "up the middle", "", "up the middle",  
                 "into the net", ""]  
topPlayerName: S. Tsitsipas  
bottomPlayerName: N. Kyrgios
```

Figure 3.2: Raw output from the computer vision module for a single tennis point.

entry contains the start frame, end frame, and court position (“top” or “bottom”) of the player who hit the shot. The frames per second (fps) and total frame count provide critical timing information about the video. The annotation start frame indicates whether an offset must be applied - if annotations begin at a frame greater than 0, we shift the shot ranges accordingly to align with the original input video. The shot classifications specify the type of each shot (e.g., Serve, Forehand, Backhand), while the shot directions describe where on the court the ball was directed (e.g., “up the middle”, “out wide”, “into the net”). Finally, the top and bottom player fields allow us to map court position to player identity, determining who hit each shot.

3.1.2 Frame-to-Time Conversion

To later assign accurate timestamps to the generated commentary, we first convert the frame ranges for each shot into time intervals. For each shot, we compute the start time as:

$$t_{\text{start}} = \frac{f + f_{\text{offset}}}{\text{fps}} \quad (3.1)$$

where f is the starting frame for that shot candidate range and f_{offset} is the annotation start frame. We then adjust the intervals to be continuous by setting the end of each interval to be just before the start of the next shot:

$$t_{\text{end}} = t_{\text{next_start}} - 0.01 \quad (3.2)$$

For the final shot, we use the ending frame of the shot range to compute t_{end} . This ensures that the commentary segments do not overlap and can be seamlessly stitched together.

3.1.3 Word Count Calculation

To guide the LLM in generating appropriately-lengthed commentary for each shot, we calculate a target word count based on the shot duration. Initially, we attempted to provide the actual timing ranges for each shot directly to the LLM and instruct it to generate commentary that fits within those time constraints. However, we found that the model struggled to produce an appropriate amount of words for a given time range despite explicit instructions, often generating commentary that was too long or too short for the available duration. To address this, we pivoted to calculating explicit word counts for each shot and instructing the model to adhere to these targets, which yielded significantly better results.

We determined empirically that a speech rate of 4 words per second represents the maximum reasonable rate for our TTS system. Thus for each shot:

$$\text{words} = \lfloor (t_{\text{end}} - t_{\text{start}}) \times 4 \rfloor \quad (3.3)$$

For intro and outro segments, we use a slower rate of 2.5 words per second to avoid rushed delivery. We automatically determine whether there is sufficient time for introductory or concluding commentary based on the amount of time before the first shot and after the last shot. If the first shot begins more than 1.5 seconds into the clip, we allocate the time before the first shot for an introductory phrase. Similarly, if more than 1.5 seconds remain after the last shot, we allocate this time for concluding commentary.

3.1.4 Combining with Tournament Metadata

The processed CV output is then combined with tournament-provided metadata. In this phase of the project, we were working primarily with points from Wimbledon 2023, and had access to the ground truth point information provided by the tournament organizers. This metadata includes contextual information such as:

- Server and non-server names
- Current set number along with set, and game scores
- Point outcome and a one-sentence description of the point
- Information about the serve (ace, serve number, serve speed) and the type of point (break point etc.)
- Excitement score for the point

Figure 3.3 shows the final transformed metadata that is used in the prompt for commentary generation.

```
Example Metadata 1

serverName: Kyrgios
nonServerName: Tsitsipas
setNum: 3
pre_set_score: [Kyrgios: 4, Tsitsipas: 2]
pre_game_score: [Kyrgios: 0, Tsitsipas: 0]
new_set_score: [Kyrgios: 4, Tsitsipas: 2]
new_game_score: [Kyrgios: 15, Tsitsipas: 0]
ace: 0
serve_number: 1
break_point: 0
break_point_won: 0
winnerOfPoint: Kyrgios
excitementOverallScore: 2
serveSpeedMPH: 115
pointDescription: Tsitsipas loses the point with a backhand unforced error.
commentary:
  intro: 9 words
  shots:
    1. [N. Kyrgios, Serve , 5 words]
    2. [S. Tsitsipas, Forehand, up the middle, 4 words]
    3. [N. Kyrgios, Backhand, up the middle, 4 words]
    4. [S. Tsitsipas, Backhand, 4 words]
    5. [N. Kyrgios, Backhand, up the middle, 5 words]
    6. [S. Tsitsipas, Backhand, into the net, 5 words]
  outro: 13 words
```

Figure 3.3: Final metadata format used for commentary generation, combining computer vision output with tournament-provided metadata.

3.2 Prompt Construction

With the metadata fully processed, we construct a prompt to pass to the large language model. In the previous system, a commentary model was finetuned on large amounts of synthetically generated data to produce realistic tennis commentary. The in-house, proprietary IBM-trained model was Instruct.Sandstone.3B. With recent advancements in LLMs, we shift this approach to use a foundation model with no finetuning. We chose to work with Llama 3.3 70B Instruct, leveraging its strong instruction-following capabilities. We set the temperature to 0.9 to encourage diverse and varied commentary generation. To construct the prompt, we perform a similar process as [5] with a few key changes. The prompt consists of several components.

3.2.1 Task Description

The prompt begins with a clear task description instructing the model to generate concise and diverse commentary for the tennis point. We provide a set of specific formatting guidelines:

- Shots are described in the order in which they occur in the format [shotHitter, shotType, shotDirection, numberOfWords]. The model should not exceed the given word count for each shot and should address the shots in consecutive order. In addition, the model should not invent any shots or extrapolate to fill any missing information.
- The model is instructed to spell out all numbers and abbreviations for alignment with the TTS system we will use later on.
- Importantly, for each commentary line, the model must indicate which shots that line corresponds to (e.g., shot 1, shot 2 or mark it as intro or outro). This phrase-to-shot mapping is critical for the timestamp generation step.

3.2.2 Excitement Modulation Instructions

The prompt includes instructions to modulate the excitement level of the commentary based on the excitementOverallScore field in the metadata. We define three levels:

- Level 1 (Low Excitement): Calm, straightforward commentary
- Level 2 (Medium Excitement): More energetic delivery with occasional exclamations
- Level 3 (High Excitement): Enthusiastic, over-the-top commentary with strong emotional language.

Section 3.4.1 describes how we determine the appropriate excitement level for each point.

3.2.3 Few-Shot Examples

To guide the model’s output format and style, we include three metadata-commentary example pairs in the prompt. These examples demonstrate the expected output format with phrase and shot correspondence markers. They also illustrate how to combine multiple shots into a single commentary phrase when appropriate, how excitement level affects the tone and language of the commentary and appropriate handling of intro and outro segments.

Figure 3.4 shows one of the example pairs included in the prompt. This example demonstrates that the model can omit the intro even when words are allocated for it, if there is no relevant contextual information to convey. It also illustrates medium excitement

(level 2), which produces energetic language with occasional exclamation points. It shows how consecutive shots can be combined into a single phrase (shots 2 and 3), with their word counts combined accordingly.

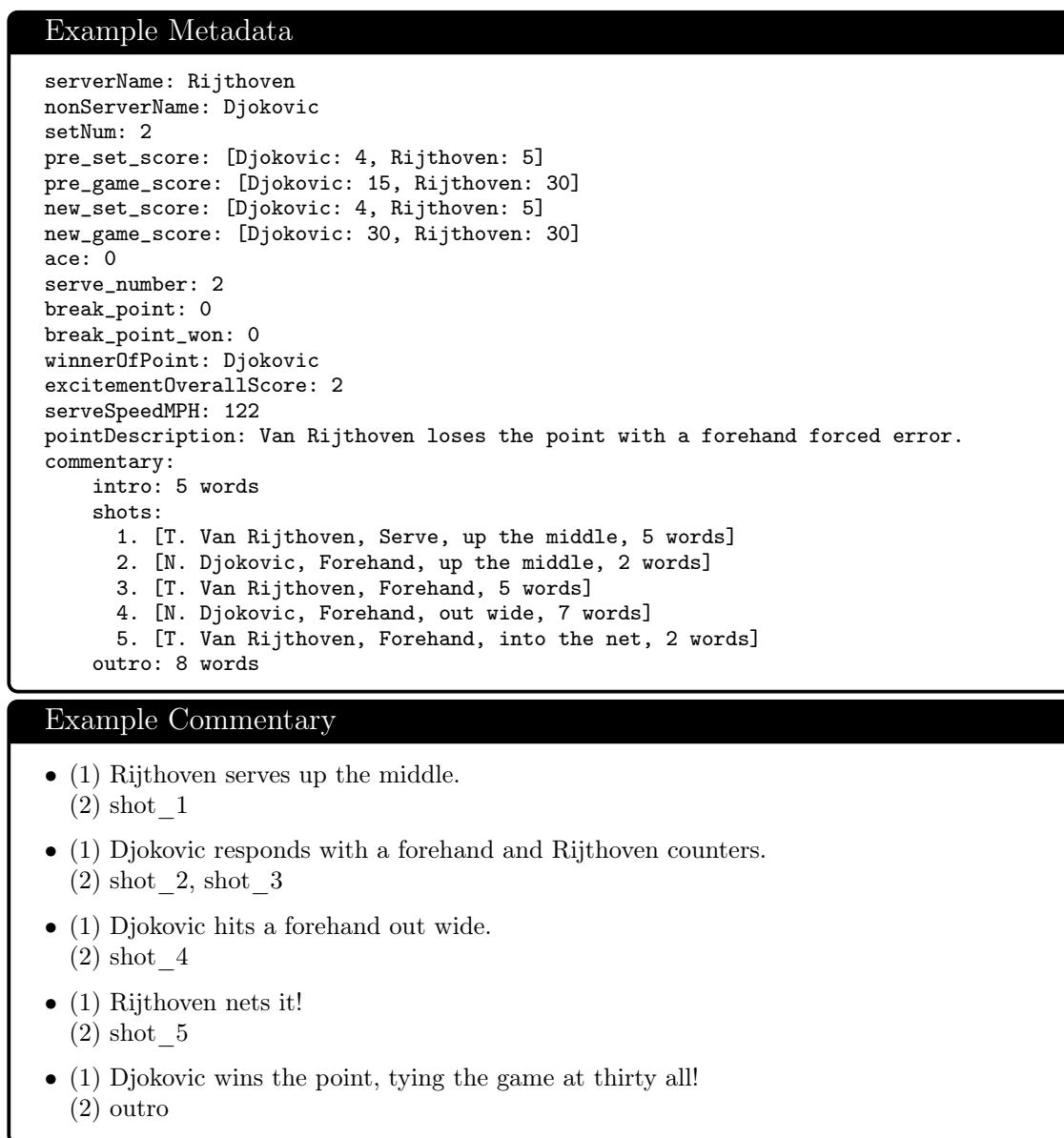


Figure 3.4: Example metadata and commentary used as a few-shot example in the prompt.

3.3 Generating Timestamps

Once we have the raw output from the LLM, we add timestamps in a post-processing step. As described in Section 3.1.3, we initially attempted to have the LLM generate timestamps

directly by including the timing ranges in the prompt and asking it to output start and end times for each phrase. However, even with chain-of-thought prompting approaches, the model struggled to produce accurate timestamps that aligned with the shot boundaries. Handling this algorithmically allows us to accurately construct time intervals and dynamically adjust speech rates based on the generated commentary.

3.3.1 Parsing the LLM Output

The first step in post-processing is parsing the raw text output from the model. The model outputs commentary in a structured format:

- (1) Commentary phrase text.
- (2) `shot_number(s)/intro/outro`

We use regular expressions to extract each phrase and its corresponding shot indices (or intro and outro markers). This parsing step produces a list of structured items, each containing a commentary phrase and a list of associated shots.

3.3.2 Timestamp Calculation Algorithm

For each parsed phrase, we calculate the start and end timestamps using the following rules.

Start Time Determination:

- If the phrase corresponds to the intro, the start time is the beginning of the intro time range
- If the phrase corresponds to only the outro, the start time is the beginning of the outro time range.
- Otherwise, the start time is the beginning of the first shot referenced by that phrase.
- To ensure continuity, if a previous phrase exists, we take the maximum of the computed start time and the previous phrase’s end time plus 0.01 seconds.

End Time Determination: We compute three candidate end times and select among them.

- Candidate 1 (Shot-based): The end of the last shot referenced by the phrase (or the outro if applicable).

- Candidate 2 (Maximum speech rate): The start time plus the phrase duration at the maximum speech rate (4 words per second for shots, 2.5 for intro/outro).
- Candidate 3 (Minimum speech rate): The start time plus the phrase duration at the minimum speech rate (2 words per second).

The selection logic is as follows:

- If Candidate 2 > Candidate 1, we use candidate 2. This handles cases where the phrase is too long to fit within the shot duration at the maximum speech rate. We extend the time to avoid unnaturally fast speech.
- Else if Candidate 3 < Candidate 1, we use Candidate 3. This handles cases where the phrase is very short. We compress the time to avoid unnaturally slow speech.
- Otherwise, we use Candidate 1, the shot-based end time.

This approach ensures that speech rates remain within the acceptable range of 2-4 words per second while respecting the natural boundaries of the tennis action when possible. The word limits enforced during prompting ensure that any necessary extensions to a shot’s time range remain close to the natural boundaries of the rally. Figure 3.5 illustrates how the shot references in the LLM output (top) are mapped to precise timestamps in the final commentary (bottom).

3.4 Modulating the Commentary with Excitement Information

Once we had the overall structure of the commentary in place, we explored how to modulate the excitement level based on metadata information. The tournament-provided metadata includes an overall excitement score computed from crowd noise levels, player gestures, and point analytics (e.g., rally length, break points, winners). This score helps us determine the appropriate energy level for the commentary.

3.4.1 Point-Level Excitement

At the point level, we map the raw excitement score (a continuous value between 0 and 1) to one of three discrete levels. We initially tried providing the raw score directly to the model, but found that it struggled to interpret the 0-1 value meaningfully. Mapping to discrete levels yielded more reliable and distinguishable commentary styles:

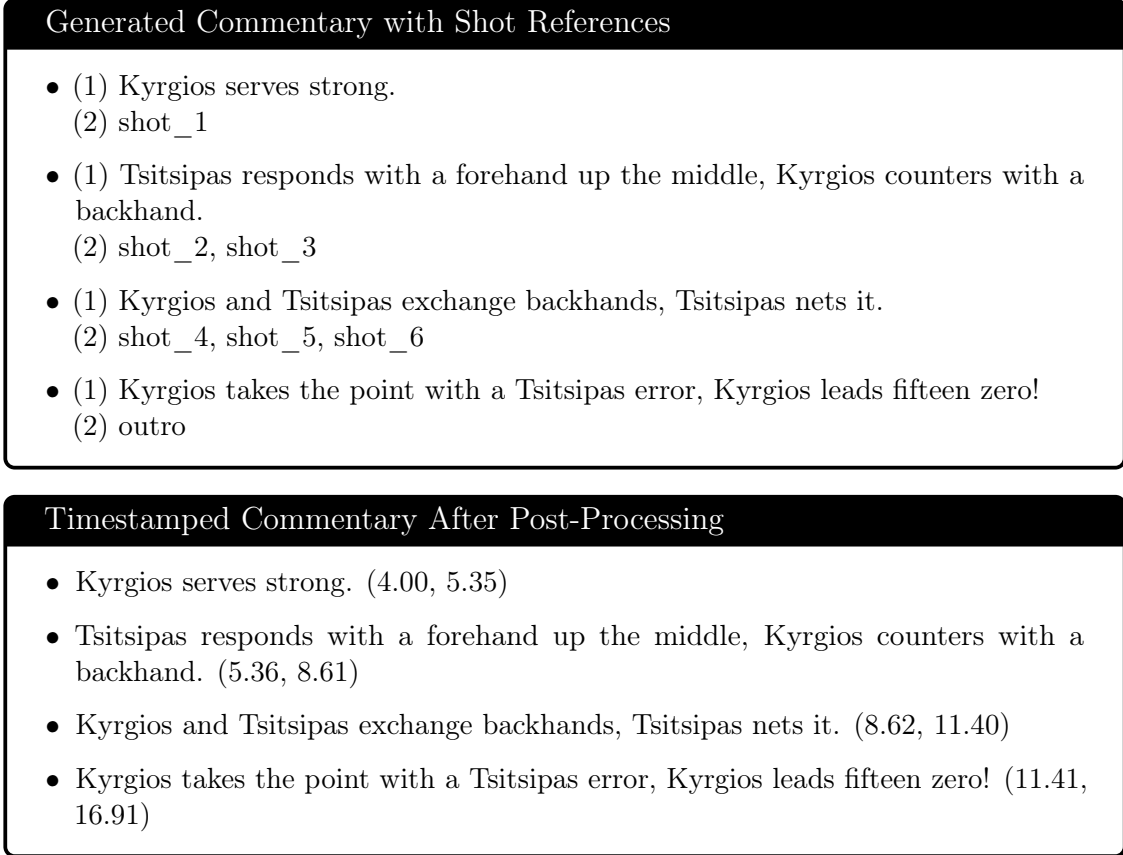


Figure 3.5: Commentary generation pipeline for the metadata in Figure 3.3. Top: raw LLM output with shot reference markers used for timestamp alignment. Bottom: final timestamped commentary after post-processing.

- Level 1 (Low): Raw score ≤ 0.4
- Level 2 (Medium): Raw score between 0.4 and 0.65
- Level 3 (High): Raw score > 0.65

This discrete level is included in the metadata passed to the LLM, which uses it to adjust the overall tone and language of the generated commentary.

3.4.2 Sentence-Level Excitement for Prosody

For TTS synthesis, we require finer-grained excitement information at the sentence level. Real sports commentary typically builds in intensity throughout a point, reaching peak excitement at the conclusion. We model this by assigning each sentence an excitement value on a 1-5 scale based on its position in the commentary and the overall point excitement.

- Overall Excitement Level 1: All sentences receive excitement value 1.

- Overall Excitement Level 2: Excitement interpolates linearly from 1 to 3 across the phrases.
- Overall Excitement Level 3: The last two phrases receive excitement value 5; earlier phrases interpolate from 2 to 4.

Additionally, sentences with excitement level 5 are automatically punctuated with exclamation marks if they do not already end with one, reinforcing the excited tone for the TTS system. We also include an outro indicator flag (0 or 1) for each phrase, which signals to the audio generation pipeline that a phrase is part of the concluding commentary. This allows the TTS system to potentially apply different prosodic treatment to outro phrases, such as slightly slower pace or different intonation patterns. A phrase is marked as outro if it addresses information beyond the shot sequence, such as point outcome.

Figure 3.6 shows the final output format for the example from Figure 3.5 with timestamps, sentence-level excitement scores and outro indicators. The tuple format accompanying each phrase is: (start_time, end_time, excitement_score, outro_indicator).

Final Commentary with Timestamps and Prosody Tags	
• Kyrgios serves strong.	(4.00, 5.35, 1, 0)
• Tsitsipas responds with a forehand up the middle, Kyrgios counters with a backhand.	(5.36, 8.61, 2, 0)
• Kyrgios and Tsitsipas exchange backhands, Tsitsipas nets it.	(8.62, 11.40, 3, 0)
• Kyrgios takes the point with a Tsitsipas error, Kyrgios leads fifteen zero!	(11.41, 16.91, 3, 1)

Figure 3.6: Final commentary output with timestamps, sentence-level excitement scores (1–5 scale), and outro indicators (1 = outro phrase, 0 = non-outro).

Figures 3.8, 3.9, and 3.10 demonstrate how the same rally produces different commentary at each excitement level. These differences in tone, word choice and punctuation are key components in producing natural-sounding commentary with appropriate prosody when combined with the TTS module.

3.5 US Open Behind the Scenes demo 2025

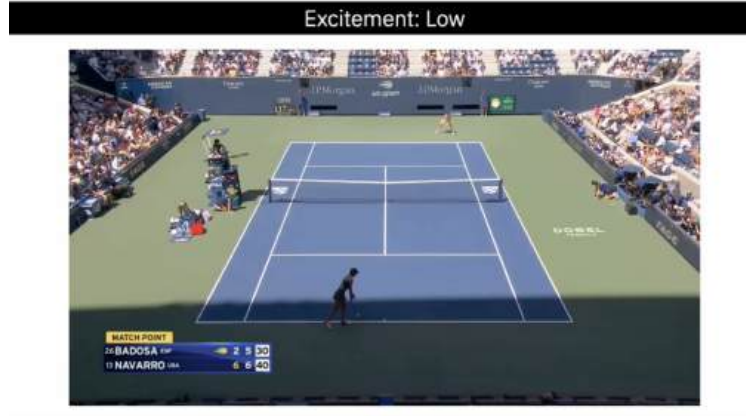
The US Open annually hosts a “Behind the Scenes” session showcasing emerging tennis technologies to media, sponsors, and industry professionals. In 2025, our excitement-driven AI sports commentary system was presented as a part of this demo [34]. Press surrounding the

demo highlighted the motivation behind this work. Major tournaments like the US Open feature over 300 matches across 15 days, making it logistically difficult and expensive to staff human announcers for every court. As Principle Scientist Rogerio Feris, one of my thesis supervisors and collaborators on this project, explained in an article: “The idea of AI-generated commentary is not to replace humans. It’s to augment humans and provide more coverage for courts that currently lack commentary” [34].

The demo highlighted the shift from monotone AI commentary to excitement-driven delivery. IBM Research Yang Zhang, another one of my collaborators on this project, notes that “instead of AI commentators that speak in a monotone level of excitement and sound unnatural to audiences, these tools can express a high excitement level just like human commentators, who get much more expressive during a very exciting rally” [34].

Figure 3.7 shows screenshots from one of the demo videos presented at the session. This video showcases the results of the end-to-end pipeline, rendering the same tennis point at different excitement levels. The contrast between the low and high excitement versions illustrates how the system modulates the textual content and prosodic delivery to match the desired tone.

Looking forward, potential extensions include allowing fans to personalize their commentary experience by selecting their preferred excitement level. The team has also received interest from researchers and clients working on other sports, such as Formula 1 racing.



(a) Low excitement commentary



(b) High excitement commentary

Figure 3.7: Screenshots from the demo video presented at the US Open 2025 Behind the Scenes session, showing the same tennis point rendered with different excitement levels. (a) Low excitement produces calm, measured delivery. (b) High excitement produces energetic, emphatic commentary with exclamation marks and expressive phrasing. The full demo video is available [here](#).

Watch Highlight 1312

Level 1: Low Excitement

Kyrgios serves, Tsitsipas responds with a forehand up the middle. (4.00, 6.50, 1, 0)
Kyrgios and Tsitsipas exchange backhands, Kyrgios up the middle. (6.51, 10.03, 1, 0)
Tsitsipas hits a backhand into the net, Kyrgios wins the point. (10.04, 15.54, 1, 0)
Kyrgios takes the point, leads fifteen zero. (15.55, 17.95, 1, 1)

Level 2: Medium Excitement

Kyrgios serves and Tsitsipas responds with a forehand. (4.00, 6.47, 1, 0)
Kyrgios and Tsitsipas exchange backhands, great rally. (6.48, 8.63, 2, 0)
Tsitsipas hits a backhand into the net, Kyrgios wins the point! (10.04, 15.54, 3, 0)
Nice point by Kyrgios he takes the lead fifteen zero! (15.55, 19.15, 3, 1)

Level 3: High Excitement

Kyrgios serves at one fifteen MPH. (4.00, 5.35, 2, 0)
Tsitsipas counters with forehand up the middle. (5.36, 7.11, 3, 0)
Kyrgios directs backhand up the middle setting up a series of backhands. (7.12, 10.12, 4, 0)
TSITSIPAS MISSES INTO THE NET! KYRGIOS TAKES THE POINT! (10.13, 15.13, 5, 1)
What a start for Kyrgios he leads fifteen zero! (15.14, 17.54, 5, 1)

Figure 3.8: Sample commentary outputs for a Kyrgios versus Tsitsipas rally under different excitement levels.

Watch Highlight 1401

Level 1: Low Excitement

Rijthoven serves up the middle. (2.12, 3.55, 1, 0)
Djokovic responds with a forehand and Rijthoven counters. (3.56, 5.63, 1, 0)
Djokovic hits out wide, Rijthoven nets a forehand and Djokovic takes the point. (5.64, 11.52, 1, 1)
Djokovic evens it up at thirty all. (11.53, 13.93, 1, 1)

Level 2: Medium Excitement

Rijthoven serves up the middle. (2.12, 3.55, 1, 0)
Djokovic responds with a forehand and Rijthoven counters. (3.56, 5.63, 2, 0)
Djokovic hits a forehand out wide and Rijthoven fails to return. (5.64, 8.14, 3, 0)
Djokovic wins the point tying the game at thirty all! (8.15, 11.75, 3, 1)

Level 3: High Excitement

Rijthoven serves up the middle. (2.12, 3.55, 2, 0)
Djokovic responds with a forehand followed by consecutive forehands from both players. (3.56, 7.43, 4, 0)
Rijthoven's forehand sails into the net and Djokovic takes the point! (7.44, 11.84, 5, 1)
UNBELIEVABLE ERROR! DJOKOVIC TIES THE GAME THIRTY ALL! (11.85, 14.65, 5, 1)

Figure 3.9: Sample commentary outputs for a Rijthoven versus Djokovic rally under different excitement levels.

Watch Highlight 1125

Level 1: Low Excitement

Andujar serves (2.00, 3.00, 1, 0)

Norrie responds with a backhand. (3.01, 4.98, 1, 0)

They exchange forehands and backhands up the middle and out wide. (4.99, 7.76, 1, 0)

Norrie loses the point with a backhand error. Andujar leads fifteen zero. (10.29, 14.09, 1, 1)

Level 2: Medium Excitement

Andujar serves and Norrie responds with a backhand. (0.64, 3.71, 1, 0)

The rally continues with forehands and backhands up the middle and out wide. (3.72, 8.28, 2, 0)

Norrie's backhand error gives Andujar the point! (8.29, 11.79, 3, 1)

Andujar takes the lead at fifteen zero! (11.80, 14.30, 3, 1)

Level 3: High Excitement

Andujar serves (2.00, 3.00, 2, 0)

Norrie responds with a backhand up the middle. (3.01, 5.01, 2, 0)

Andujar counters with a forehand and Norrie hits backhand out wide. (5.02, 7.96, 3, 0)

Andujar fires another forehand, Norrie's backhand goes out wide again! (7.97, 10.47, 4, 0)

ANDUJAR TAKES THE POINT WITH NORRIE'S ERROR!" (10.48, 12.75, 5, 1)

"What a play Andujar leads fifteen zero!" (14.05, 17.05, 5, 1)

Figure 3.10: Sample commentary outputs for a Andujar versus Norrie rally under different excitement levels.

Chapter 4

Unifying Visual Understanding and Commentary Generation

As described in Chapter 2, StreamingVLM provides a promising foundation for unified visual understanding and commentary generation for tennis matches. The model was trained on over 4,000 hours of sports commentary covering basketball, soccer, ice hockey, baseball, and American football, but notably not tennis. To adapt StreamingVLM to the tennis domain, we aim to build a high-quality dataset of tennis video clips paired with timestamped commentary suitable for fine-tuning. For training, StreamingVLM splits long video streams into manageable chunks. For our tennis data, we sought to produce segments of manageable duration with semantically coherent boundaries. Each point of the tennis match forms a self-contained action sequence from serve to point conclusion, making it an ideal chunk boundary.

We explore two complementary methods for generating this type of training data, each with distinct strengths and limitations.

4.1 Method 1: Leveraging the Existing Commentary Pipeline

Our first approach utilizes the computer vision module and LLM-based commentary generation system described in Chapter 3. As shown in Figure 3.1, the full pipeline extends through ProsodyLM to generate spoken commentary. However, for data generation, we only execute the pipeline through the post-processing stage, extracting the timestamped commentary without the prosody tags. By running this partial pipeline on tennis point clips, we generate training examples with shot-by-shot commentary in our desired play-by-play style.

4.1.1 Strengths and Limitations

This method offers several advantages. The generated commentary is precisely aligned with the visual content, as it is derived from computer vision analysis of the actual shots. The commentary follows our desired play-by-play format with shot descriptions, player names, and score updates. When tournament-provided metadata is available, we can adjust the commentary excitement level to create varied training examples.

However, this approach has significant drawbacks. Due to the latency of running the computer vision module and LLM inference for each clip, this method is not feasible for generating large-scale training data. Also, if the shot classification or direction detection from the CV module is inaccurate, the resulting commentary will be incorrect, making it unsuitable for training. These training examples were generated from isolated point clips from Wimbledon 2023. Unlike StreamingVLM’s original training data, which consists of continuous match footage with accumulated context from previous commentary, these clips are standalone points without surrounding match context. This structural difference means our Method 1 examples lack the “previous” commentary field that StreamingVLM uses to maintain continuity across a broadcast. The isolated clips do not provide the continuous match context that StreamingVLM expects, potentially limiting the model’s ability to learn contextual commentary patterns.

4.1.2 Dataset Statistics

Using Method 1, we generated a total of 148 commentary examples from Wimbledon 2023 match clips. Figure 4.1 shows the distribution across excitement levels.

Excitement Level	Count	Percentage
Level 1 (Low)	67	45.3%
Level 2 (Medium)	48	32.4%
Level 3 (High)	33	22.3%
Total	148	100%

Figure 4.1: Distribution of Method 1 training examples by excitement level.

The output from Method 1 is formatted for StreamingVLM’s training pipeline as JSONL, where each line contains a conversation with video input and timestamped commentary output. Following the StreamingVLM paper, we represent temporal alignment at the word level, with each word in the commentary paired with precise start and end timestamps, allowing the model to learn precise synchronization between visual events and linguistic descriptions [25]. Figure 4.2 shows an example training instance.

StreamingVLM Training Format for Method 1

```
[
  {"role": "user", "content": [
    {"type": "video",
     "video": "1408_8_db.mp4",
     "video_start": 2.16,
     "video_end": 11.18},
    {"type": "text",
     "text": "Commentate on this match"}
  ]},
  {"role": "assistant", "content": [{"type": "text_stream", "text_stream": [
    [2.16, 2.5, "Djokovic"], [2.5, 2.84, "serves"],
    [2.84, 3.18, "out"], [3.18, 3.52, "wide."],
    [3.53, 3.98, "What"], [3.98, 4.43, "a"],
    [4.43, 4.88, "serve!"], [4.88, 5.33, "Djokovic"],
    [5.33, 5.78, "wins"], [5.78, 6.23, "the"],
    [6.23, 6.68, "point"], [6.68, 7.13, "with"],
    [7.13, 7.58, "an"], [7.58, 8.03, "ace!"],
    [8.03, 8.48, "Djokovic"], [8.48, 8.93, "leads"],
    [8.93, 9.38, "forty"], [9.38, 9.83, "love"],
    [9.83, 10.28, "in"], [10.28, 10.73, "the"],
    [10.73, 11.18, "game!"]]
  ]}
]
```

Figure 4.2: Example training instance for Method 1 data in StreamingVLM format. Each commentary segment has the format [start_time, end_time, word].

4.2 Method 2: Cleaning Transcribed Streaming Commentary

Our second approach develops a scalable pipeline for processing existing tennis commentary from YouTube streams. We use commentary from Cam Williams, a popular tennis streamer who provides live play-by-play commentary over full match broadcasts.

4.2.1 Strengths and Limitations

This method addresses the scalability limitations of Method 1. Once the pipeline is established, it can be applied to process many matches relatively quickly. The commentary captures natural prosodic variation in an authentic commentator style, including varied phrasing and reactions that we had to emulate with Method 1. In addition, the transcripts include contextual commentary, player analysis, and emotional reactions that go beyond pure shot

descriptions.

However, there is significant noise in the transcripts. Some segments contain non-tennis content (commentary on what people are saying in the streaming chat, personal anecdotes, etc.) that must be filtered out. In some cases, there is little to no usable and informative commentary for a specific point. This limits the amount of training examples we can extract from a given match. In addition, automatic speech recognition (ASR) introduces transcription errors, which can degrade the quality of the commentary if not handled properly.

4.2.2 Pipeline Overview

The Method 2 pipeline consists of four stages, illustrated in Figure 4.3

1. **Score Detection via Optical Character Recognition:** In order to split a full match video into smaller clips for fine-tuning, we extract point boundaries and score metadata from the video scoreboard
2. **Audio Transcription:** We transcribe the commentary audio in 5-second chunks using NVIDIA’s Parakeet ASR Model.
3. **Audio-Video Alignment:** We align transcribed segments with detected point boundaries using an offset between the streaming timeline and the match video timeline.
4. **LLM-based Cleaning:** Lastly, we use an LLM to clean and restructure the raw transcripts into proper commentary format.

4.2.3 Stage 1: Score Detection via Optical Character Recognition

To segment full match videos into individual points, we use optical character recognition (OCR) to detect score changes on the broadcast scoreboard. We define regions of interest (ROIs) for the match score, set score, game score and server indicator. The green column indicates the match score (sets won), the purple column indicates the set score (games won), and the white column displays the current game score. The purple arrow on the left indicates which player is serving. Player 1 corresponds to the top row and player 2 to the bottom row. Actual player names are known from the match metadata and substituted in during later processing.

We sample frames at 5-second intervals and apply OCR to extract the current scores. We determined that 5-second sampling was appropriate for efficiency and was sufficient to capture all score changes, as points in professional tennis typically last longer than this

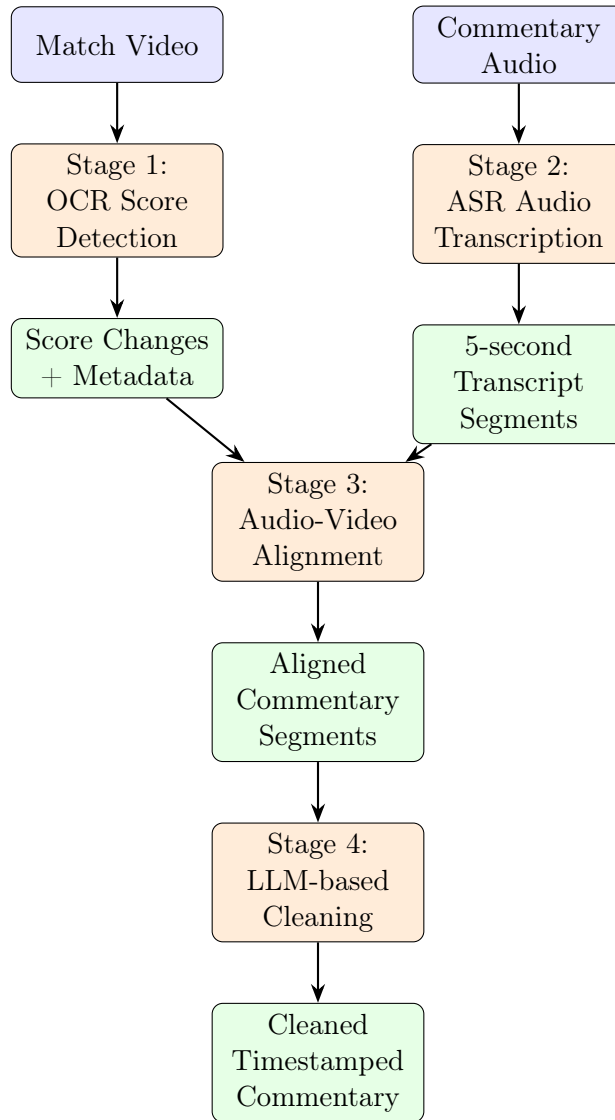


Figure 4.3: Method 2 pipeline for processing tennis commentary. The pipeline takes match video and commentary audio as inputs, extracts score changes via OCR (Stage 1), transcribes audio in 5-second chunks (Stage 2), aligns the two streams using a determined offset (Stage 3), and cleans the transcripts using an LLM (Stage 4).

interval, and the scoreboard remains stable for several seconds after each point concludes. The system tracks score progression and validates detected values against tennis scoring rules. For example, game scores must be in $\{0, 15, 30, 40, \text{AD}\}$, set scores must be valid integers and both must progress logically. When a score change is detected, we record the timestamp along with the full score state and the server. This produces a list of score change events that define point boundaries. Figure 4.4 shows the scoreboard region extracted from the broadcast along with the structured output produced by the OCR pipeline.

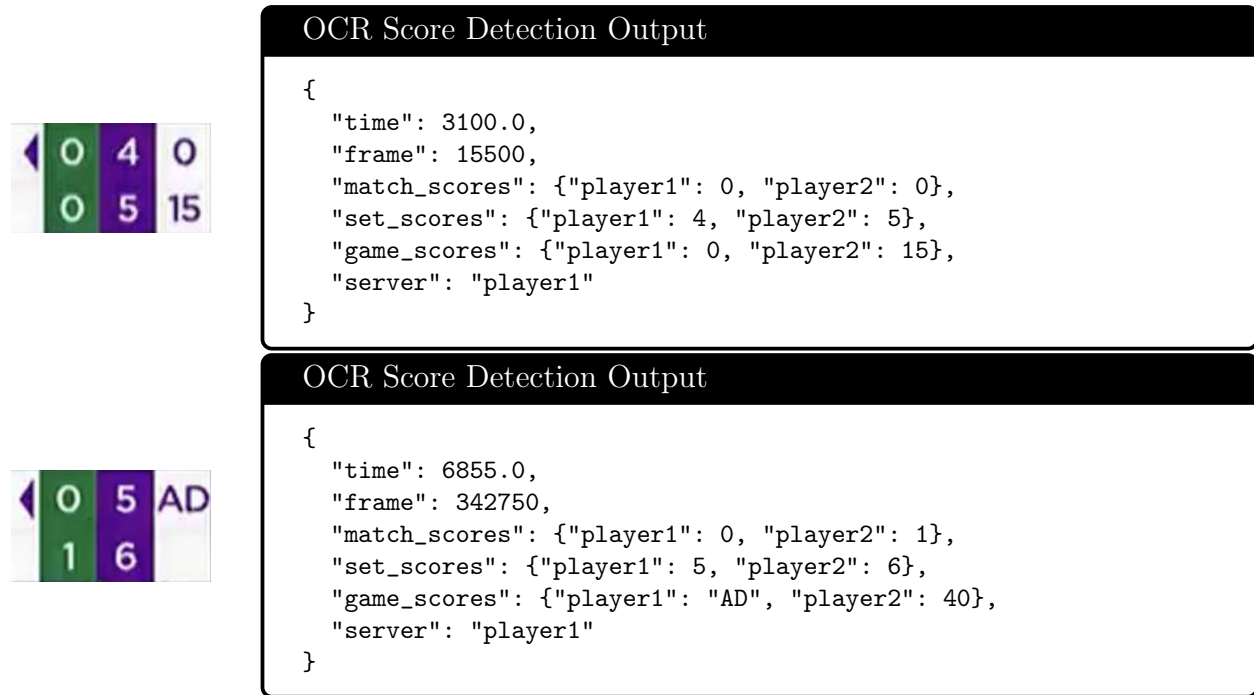


Figure 4.4: Score detection via OCR. Left: the scoreboard region of interest extracted from the broadcast. Right: the structured output produced by the OCR pipeline, capturing the full score state at this timestamp. For the bottom point, player 2’s previous score was 40, so that is propagated even though it’s not on the current scoreboard.

4.2.4 Stage 2: Audio Transcription

We extract the audio track from the YouTube stream and transcribe it using one of NVIDIA’s Parakeet ASR models, specifically `parakeet-tdt-0.6b-v2` [35]. NVIDIA Parakeet is a family of open-source, high-performance ASR models built on the NeMo framework, and `parakeet-tdt-0.6b-v2` is designed for high-quality English transcription with punctuation and capitalization.

The audio is processed in 5-second chunks, producing a list of segments with start time, end time, and transcribed text. Figure 4.5 shows an excerpt of transcribed commentary for the duration of a point.

Audio Transcription Output (5-second segments)

```
{"start_s": 4215.0,
"end_s": 4220.0,
"text": "And this is the same thing."}

{"start_s": 4220.0,
"end_s": 4225.0,
"text": "Serves fault. Second serve."}

{"start_s": 4225.0,
"end_s": 4230.0,
"text": "Second serve."}

{"start_s": 4230.0,
"end_s": 4235.0,
"text": "Down to the T, the backhand of Djokovic, slice Mussetti."}

{"start_s": 4235.0,
"end_s": 4240.0,
"text": "Lafona Djokovic. Musetti down on the middle. Djokovic goes cross-court
with the backhand."}

{"start_s": 4240.0,
"end_s": 4245.0,
"text": "The back of Djokovic. The back of Musetti. If he smacks it down the line,
Djokovic is there."}

{"start_s": 4245.0,
"end_s": 4250.0,
"text": "There's Djokovic. It's a forehand cross court. He said he gets it. Djokovic
backhand."}

{"start_s": 4250.0,
"end_s": 4255.0,
"text": "Back into his own. Djokovic cross court again. Musetti again at the back of
the Djokovic."}

{"start_s": 4255.0,
"end_s": 4260.0,
"text": "Stretch Joker to control back across court winner love 15."}

{"start_s": 4260.0,
"end_s": 4265.0,
"text": "Too good."}

{"start_s": 4265.0,
"end_s": 4270.0,
"text": "Way too good. Love fifteen. Is that tea or coffee? Tea."}

{"start_s": 4270.0,
"end_s": 4275.0,
"text": "T always tea."}
```

Figure 4.5: Example output from the audio transcription stage. Each segment contains the start and end timestamps (in seconds) and the transcribed text. Note the mix of tennis commentary and non-tennis conversation, as well as ASR errors in player names (e.g., “Mussetti” instead of “Musetti”, “Joker”).

4.2.5 Stage 3: Audio-Video Alignment

The YouTube commentary stream is offset from the original match video. The streamer begins streaming before the match begins, leading to an offset between the transcribed commentary and determined point boundaries. We determine this offset by identifying the start of the match in the commentary audio and aligning that with the start of the match in the match video.

Using this offset, we align each detected point (from Stage 1) with its corresponding commentary segments (from Stage 2). The alignment produces a structured record for each point containing:

1. The rally start and end times (in the video timeline)
2. The corresponding audio start and end times (in the commentary timeline)
3. Metadata about the point including the game, set and match scores before and after the point along with the server and non-server names and the winner of the point
4. Raw commentary segments that correspond to this point

The alignment script uses winner determination logic that checks for score changes at the match, set and game levels to accurately identify the winner of each point. Figure 4.6 shows an example of this alignment with an accurate determination of the point winner.

4.2.6 Stage 4: LLM-Based Commentary Cleaning

As shown in Figures 4.5 and 4.6, the raw transcripts contain ASR errors, non-tennis content and unstructured phrasing. We use an LLM (Gemini 2.5) to clean and restructure each point’s commentary into the target format. The cleaning prompt instructs the model to transform the messy play-by-play commentary into tuples of the form (start_time, end_time, commentary), using only details that already exist in the provided transcript segments without inventing new shots, scores or events. The model is instructed to stay true to the timeline in the provided commentary segments and adjust the time ranges for each sentence to maintain an average word rate between 2-4 words per second. The prompt emphasizes concise, energetic phrasing that mirrors the original tone without copying ASR errors. If none of the transcript text relates to players, tennis action, or scoring, the model returns “NO COMMENTARY AVAILABLE”, allowing us to filter out points with insufficient commentary content.

The prompt includes several examples demonstrating the desired output format, covering different types of rally descriptions and player interactions. This few-shot prompting approach helps the LLM understand the target style and structure. Here is an example:

Aligned Point Data

```
{ "rally_start": 3045.0, "rally_end": 3105.0,
  "audio_start": 4215.0, "audio_end": 4275.0,
  "serverName": "Musetti", "nonServerName": "Djokovic",
  "setNum": 1,
  "pre_match_score": {"Musetti": "0", "Djokovic": "0"},
  "pre_set_score": {"Musetti": "4", "Djokovic": "5"},
  "pre_game_score": {"Musetti": "0", "Djokovic": "0"},
  "new_match_score": {"Musetti": "0", "Djokovic": "0"},
  "new_set_score": {"Musetti": "4", "Djokovic": "5"},
  "new_game_score": {"Musetti": "0", "Djokovic": "15"},
  "winnerOfPoint": "Djokovic",
  "commentary_segments": [
    (3045.00, 3050.00, "And this is the same thing."),
    (3050.00, 3055.00, "Serves fault. Second serve."),
    (3055.00, 3060.00, "Second serve."),
    (3060.00, 3065.00, "Down to the T, the backhand of Djokovic, slice Mussetti."),
    (3065.00, 3070.00, "Lafona Djokovic. Musetti down on the middle..."),
    (3070.00, 3075.00, "The back of Djokovic. The back of Musetti..."),
    (3075.00, 3080.00, "There's Djokovic. Forehand cross court..."),
    (3080.00, 3085.00, "Djokovic cross court again. Musetti at the back..."),
    (3085.00, 3090.00, "Stretch Joker to control back across court winner love 15."),
    (3090.00, 3095.00, "Too good."),
    (3095.00, 3100.00, "Way too good. Love fifteen. Is that tea or coffee? Tea."),
    (3100.00, 3105.00, "T always tea.")
  ]
}
```

Figure 4.6: Aligned output for a single point, combining metadata from OCR score detection with the raw transcript segments from Figure 4.5. The rally timestamps correspond to the video timeline while the audio start and end times indicate the original position in the commentary stream; the offset between them is used to align the two sources. Commentary segment timestamps have been converted to the video timeline.

- (1002.0, 1003.0, 'Wawrinka serves. '), (1004.4, 1006.4, 'Djokovic responds with a forehand up the middle. '), (1006.41, 1008.41, 'Djokovic loses the point with a forehand error! '), (1008.42 , 1012.01, 'That brings the score to fifteen all. ')

Finally, we provide the aligned point data outlined in Figure 4.6 to conclude the prompt. We found that providing both the timestamps and the existing commentary for editing yields much better results than asking the model to generate new commentary to fill a given time segment. Figure 4.7 shows the output of this cleaning procedure for the point from Figure 4.6.

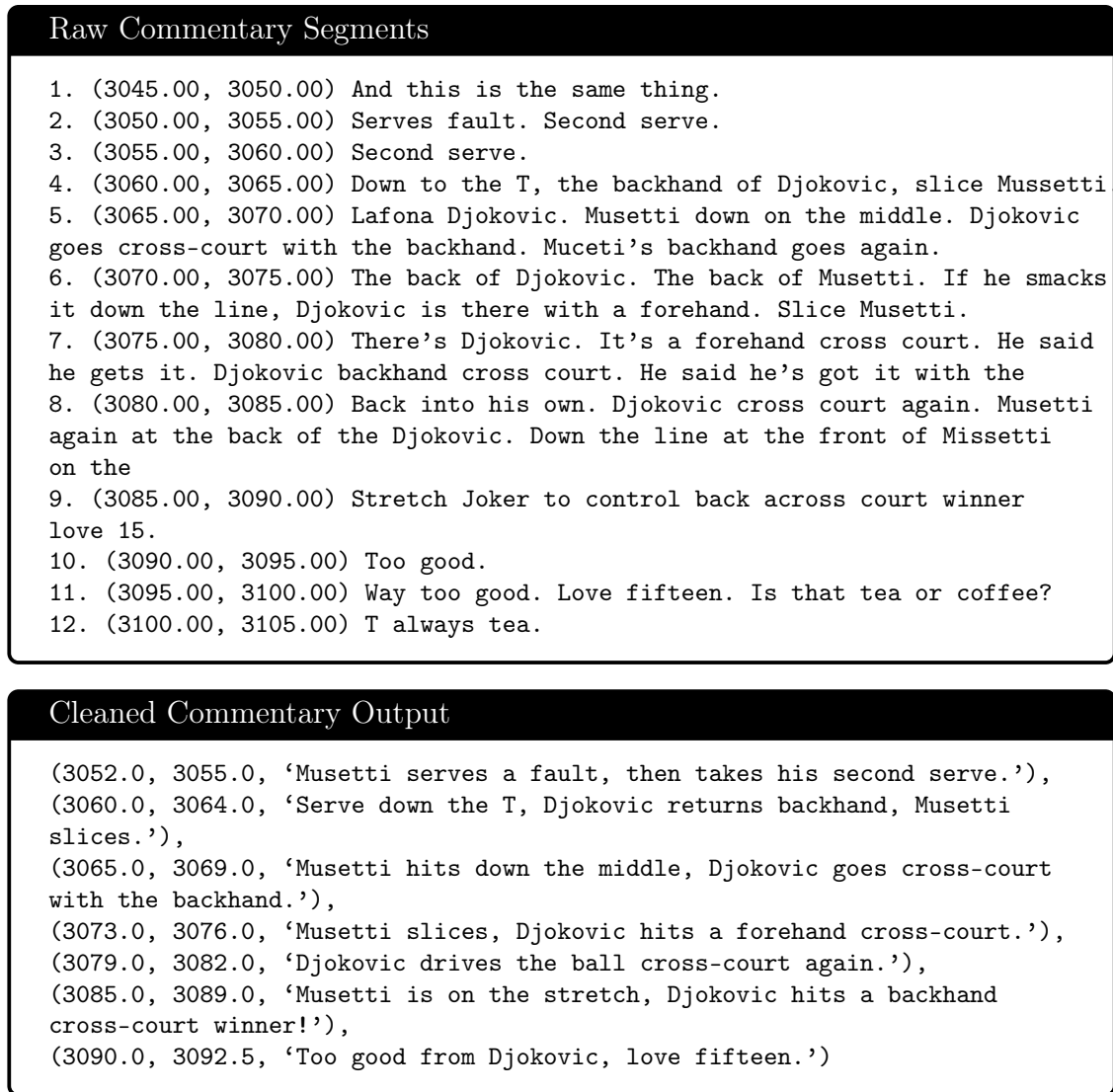


Figure 4.7: Example of raw transcription (top) being cleaned into structured commentary (bottom). The LLM removes non-tennis content, fixes ASR errors in player names, and restructures the phrasing while adhering to the original timing.

Once the commentary has been cleaned, we format it for StreamingVLM’s training pipeline. Figure 4.8 shows the final training example structure for Method 2 data. Crucially, because Method 2 processes continuous match footage rather than isolated point clips, we can populate the “previous” field with commentary from earlier points in the match. This gives the model important contextual information about the facts and narrative arc of the match. The assistant response contains the timestamped commentary, where each tuple specifies the start time, end time, and commentary text aligned to the video timeline. As with the Method 1 data, we follow the StreamingVLM format, converting phrase-level timestamps by diving each phrase’s time range equally among its constituent words.

Final Training Example for StreamingVLM

```
{
  "role": "user", "content": [
    {
      "type": "video", "video": "MS602.mp4",
      "video_start": 3052.0, "video_end": 3092.5},
    {
      "type": "text", "text":
      "Commentate on this match",
      "previous": "Djokovic serves a fault. He hits a second serve. Musetti sends
        a slice to Djokovic. Musetti hits a drop shot winner. Djokovic doesn't
        move! The score is fifteen all. Djokovic hits a second serve. They
        exchange backhands, Musetti slicing the ball...."}
  ],
  "role": "assistant", "content": [
    {
      "type": "text_stream", "text_stream": [
        [[3052.0, 3052.33, "Musetti"], [3052.33, 3052.67, "serves"],
        [3052.67, 3053.0, "a"], [3053.0, 3053.33, "fault,"],
        [3053.33, 3053.67, "then"], [3053.67, 3054.0, "takes"],
        [3054.0, 3054.33, "his"], [3054.33, 3054.67, "second"],
        [3054.67, 3055.0, "serve."], [3060.0, 3060.44, "Serve"],
        [3060.44, 3060.89, "down"], [3060.89, 3061.33, "the"],
        [3061.33, 3061.78, "T,"], [3061.78, 3062.22, "Djokovic"],
        [3062.22, 3062.67, "returns"], [3062.67, 3063.11, "backhand,"],
        [3063.11, 3063.56, "Musetti"], [3063.56, 3064.0, "slices."],
        [3065.0, 3065.36, "Musetti"], [3065.36, 3065.73, "hits"],
        [3065.73, 3066.09, "down"], [3066.09, 3066.45, "the"],
        [3066.45, 3066.82, "middle,"], [3066.82, 3067.18, "Djokovic"],
        [3067.18, 3067.55, "goes"], [3067.55, 3067.91, "cross-court"],
        [3067.91, 3068.27, "with"], [3068.27, 3068.64, "the"],
        [3068.64, 3069.0, "backhand."], [3073.0, 3073.43, "Musetti"],
        [3073.43, 3073.86, "slices,"], [3073.86, 3074.29, "Djokovic"],
        [3074.29, 3074.71, "hits"], [3074.71, 3075.14, "a"],
        [3075.14, 3075.57, "forehand"], [3075.57, 3076.0, "cross-court."],
        [3079.0, 3079.5, "Djokovic"], [3079.5, 3080.0, "drives"],
        [3080.0, 3080.5, "the"], [3080.5, 3081.0, "ball"],
        [3081.0, 3081.5, "cross-court"], [3081.5, 3082.0, "again."],
        [3085.0, 3085.36, "Musetti"], [3085.36, 3085.73, "is"],
        [3085.73, 3086.09, "on"], [3086.09, 3086.45, "the"],
        [3086.45, 3086.82, "stretch,"], [3086.82, 3087.18, "Djokovic"],
        [3087.18, 3087.55, "hits"], [3087.55, 3087.91, "a"],
        [3087.91, 3088.27, "backhand"], [3088.27, 3088.64, "cross-court"],
        [3088.64, 3089.0, "winner!"], [3090.0, 3090.42, "Too"],
        [3090.42, 3090.83, "good"], [3090.83, 3091.25, "from"],
        [3091.25, 3091.67, "Djokovic,"], [3091.67, 3092.08, "love"],
        [3092.08, 3092.5, "fifteen."]]]]]}
  ]
}
```

Figure 4.8: Final training example for Method 2 in StreamingVLM format. The user turn contains the video segment and a “previous” field with commentary context from earlier in the match. The assistant turn contains timestamped commentary as a `text_stream`, where each tuple is `[start_time, end_time, word]`. This format aligns with StreamingVLM’s training pipeline.

4.2.7 Initial Scaling

To work towards building a sufficiently large training dataset, we refined the pipeline on the Wimbledon 2024 match between Lorenzo Musetti and Novak Djokovic (Match 0). We then applied the Method 2 pipeline to 10 additional matches from Wimbledon 2024, all with accompanying streaming commentary. Figure 4.9 summarizes the results of this initial scaling effort.

Match	Good Commentary	No Commentary	Total Points	Yield (%)
Match 0	196	15	211	92.9
Match 1	115	63	178	64.6
Match 2	146	47	193	75.6
Match 3	201	46	247	81.4
Match 4	134	25	159	84.3
Match 5	225	54	279	80.6
Match 6	267	52	319	83.7
Match 7	146	38	184	79.3
Match 8	251	35	286	87.8
Match 9	200	56	256	78.1
Match 10	150	147	297	50.5
Total	2,031	578	2,609	77.8

Figure 4.9: Method 2 pipeline results across 11 Wimbledon 2024 matches. “Good Commentary” indicates points where the LLM cleaning stage produced usable training examples. “No Commentary” indicates points where the raw transcript contained insufficient tennis-related content (e.g., chat interactions, off-topic discussion) and the LLM returned NO COMMENTARY AVAILABLE.

The pipeline successfully extracted 2,301 usable training examples from the 11 matches, achieving an overall yield of 77.8%. The yield varied considerably across matches, ranging from 50.5% (Match 10) to 92.9% (Match 0). This variation reflects differences in the streamer’s commentary during each match. Some matches featured more extensive play-by-play coverage, while others included more discussion of chat messages, other matches, or other non-tennis content that was filtered out during the LLM cleaning stage. The low yields for Match 1 and Match 10 can be partially attributed to the streamer commentating on multiple matches concurrently during portions of those streams.

4.2.8 Method 2 Commentary Examples

To illustrate the range of commentary produced by the Method 2 pipeline, we manually categorized a sample of cleaned outputs into three excitement levels based on the content and tone of the commentary. These categorizations are subjective assessments rather than

qualitative classifications based on metadata, because we don’t have tournament metadata for those points. However, these examples demonstrate that the natural variation in the source commentary is preserved through the cleaning process, providing diverse training examples that span calm, routine points to dramatic, highlight-worthy rallies. Figures 4.10, 4.11 and 4.12 illustrate the range of commentary tones preserved through the cleaning pipeline. Low excitement examples (Figure 4.10) feature straightforward, factual descriptions of routine points with minimal expressive language. Medium excitement examples (Figure 4.11) capture notable shots and important game situations, with occasional expressive phrasing. High excitement examples (Figure 4.12) contain emphatic language, exclamation marks, and commentary on the tone of the point with phrases like “Ridiculous shot!” and “That was close”. This natural variation in the source commentary provides the model with diverse training examples spanning the full spectrum of match intensity.

Low Excitement Commentary Examples from Method 2

Example 1
(2689.00, 2691.00, “At forty-fifteen, Sinner serves.”) (2706.00, 2709.50, “Shelton’s return goes out, and Sinner holds serve.”) (2710.50, 2713.50, “Sinner now leads three-one in the second set.”)
Example 2
(542.00, 546.00, “Musetti serves to Fritz’s backhand and hits a beautiful drop shot winner.”) (546.01, 549.01, “Musetti holds, leveling the set at one game apiece.”)
Example 3
(4795.0, 4798.5, “Djokovic serves, hits a backhand, and Musetti responds with a slice.”) (4800.0, 4804.0, “Djokovic drives down the line, forcing Musetti to stretch for a backhand.”) (4806.0, 4811.0, “Musetti keeps it in play, but Djokovic puts away a crosscourt smash.”) (4811.1, 4814.1, “That brings the score to fifteen all.”)
Example 4
(6180.0, 6184.0, “Djokovic hits a slice return, Musetti counters, and Djokovic slices again.”) (6184.1, 6187.1, “Djokovic hits a forehand into the net.”) (6187.2, 6190.7, “Musetti wins the game, leveling the set at five all.”)
Example 5
(8198.00, 8201.51, “Zverev’s return goes out. Fifteen-love to Fritz.”)

Figure 4.10: Examples of cleaned commentary with low excitement characteristics: factual descriptions, minimal expressive language, and routine point outcomes.

Medium Excitement Commentary Examples from Method 2

Example 1

(4308.00, 4309.50, "Djokovic leads fifteen love."),
(4327.50, 4330.00, "Djokovic faults on the first serve."),
(4347.50, 4350.50, "Djokovic trades backhands with Fearnley from the baseline."),
(4352.50, 4356.00, "Djokovic rips a forehand crosscourt; Fearnley stretches to defend."),
(4357.00, 4359.50, "Djokovic attempts a slice but sends it long!"),
(4359.51, 4361.51, "Fearnley wins the point, fifteen all.")

Example 2

(6815.00, 6819.00, "Fritz returns; Musetti meets him at the net with a volley."),
(6819.01, 6823.51, "Fritz scrambles to get it back, but his slice lands wide!"),
(6823.52, 6826.52, "Musetti takes the point, leading forty-thirty.")

Example 3

(7980.0, 7983.0, "Djokovic fires a serve to the forehand.")
(7985.0, 7989.0, "Musetti slices the backhand, Djokovic responds, and Musetti slices again.")
(7991.0, 7995.0, "Musetti hits a slice that clips the tape and falls over! ")
(7996.0, 7999.0, "Musetti takes the point, love fifteen.")

Example 4

(785.00, 787.00, "Medvedev with a second serve."),
(790.00, 792.80, "Sinner on the attack puts away a smash!"),
(792.81, 795.81, "Sinner wins the point, making it forty-fifteen.")

Example 5

(7680.0, 7683.0, "Djokovic is moving fine and looking aggressive.")
(7687.5, 7689.5, "Musetti puts up a lob.")
(7690.0, 7694.5, "Djokovic finds an aggressive position, hits a smash, and puts it away!")
(7695.0, 7697.0, "The score is now deuce.")

Example 6

(875.00, 877.00, "Paul serves."),
(877.01, 880.01, "Alcaraz returns, but Paul's forehand goes long!"),
(880.50, 883.50, "Advantage Alcaraz, another break point.")

Figure 4.11: Examples of cleaned commentary with medium excitement characteristics: notable shots, important game situations, and some expressive phrasing.

High Excitement Commentary Examples from Method 2

Example 1

(442.0, 444.0, "Djokovic hits a second serve.")
(450.0, 454.0, "They exchange backhands, Musetti slicing the ball.")
(456.0, 460.0, "Djokovic drives down the line, Musetti defends with a slice.")
(462.0, 466.0, "Musetti keeps the ball low with another backhand slice.")
(469.0, 474.0, "Djokovic strikes a forehand inside out, Musetti returns.")
(475.0, 479.0, "Musetti attempts a drop volley, Djokovic stretches for a backhand winner!")
(480.0, 483.0, "Djokovic leads thirty fifteen!")
(485.0, 487.0, "Ridiculous shot! Djokovic wins the point.")

Example 2

(438.15, 441.15, "Advantage Paul, serving for the opening game."),
(446.88, 449.88, "Paul with a shot down the line, Alcaraz puts it in the net!"),
(449.89, 452.39, "Paul takes the first game!")

Example 3

(8481.00, 8485.00, "Set point for Musetti at forty-fifteen, but his first serve is a fault."),
(8499.00, 8504.00, "Fritz returns the second serve; after a Musetti slice, Fritz goes down the line and loses the point!"),
(8505.50, 8509.50, "And that's the set for Musetti, who now leads two sets to one!")

Example 4

(7005.0, 7009.0, "Djokovic leads three love with a double break.")
(7030.0, 7034.0, "Djokovic hits a fault on the first serve.")
(7040.0, 7043.0, "He steps up for the second serve.")
(7055.0, 7059.0, "Musetti hits a backhand, and Djokovic responds.")
(7060.0, 7064.0, "Musetti strikes a forehand crosscourt, Djokovic gets it back.")
(7065.0, 7070.0, "Musetti slices, Djokovic hits a forehand, and Musetti fires a winner!")
(7070.1, 7074.1, "That was close! Musetti gets one back, trailing one three!")

Example 5

(280.00, 282.50, "Love-forty, break point for Djokovic."),
(290.00, 291.50, "Rune serves a fault."),
(300.00, 303.50, "Djokovic returns, and Rune hits an unforced error into the net!"),
(303.51, 306.51, "Djokovic breaks! He takes an early two-love lead.")

Example 6

(150.00, 153.00, "Break point for Sinner, Medvedev serves."),
(170.00, 174.50, "A backhand-to-backhand rally, Medvedev stretches to return."),
(175.00, 179.00, "Sinner draws him forward, forcing a forehand error from Medvedev!"),
(180.00, 183.00, "Sinner wins the point, bringing the score to deuce!")

Figure 4.12: Examples of cleaned commentary with high excitement characteristics: exclamation marks, emphatic language and dramatic match situations.

4.3 Initial Fine-Tuning Results

We combine the Method 1 and Method 2 data to form a complete dataset of over 2,100 examples. Method 1 provides clean, well-structured examples that establish the desired commentary format, while Method 2 contributes natural variation and authentic commentator style. We split the combined dataset 90/10 into training and validation sets, yielding approximately 1900 training examples and 210 validation examples. Using this data, we conducted initial fine-tuning experiments on StreamingVLM to evaluate the model’s ability to learn tennis-specific commentary generation. We experimented with fine-tuning from StreamingVLM’s weights along with the Qwen-2.5-VL-7B Instruct weights, which StreamingVLM was built on top of. Due to the relatively small size of our current tennis dataset, fine-tuning from the StreamingVLM weights achieves more natural commentary cadence because of its exposure to that type of data.

4.3.1 Learning Rate Experiments

We fine-tuned StreamingVLM for 2 epochs using four different learning rates: 1×10^{-6} , 2×10^{-6} , 1×10^{-5} , and 2×10^{-5} . Training was conducted on 4 GPUs with a per-GPU batch size of 1 and gradient accumulation over 8 steps. This configuration results in just over 60 steps per epoch and 122 total optimizer steps. Figure 4.13 shows the training loss curves for each configuration, while Figure 4.14 shows the corresponding validation loss curves.

The loss curves reveal several key observations. First, the higher learning rates (1×10^{-5} and 2×10^{-5}) achieve lower final loss values on both training and validation sets. The validation loss curves show that these configurations reach approximately 0.85 by step 120, compared to 0.89 for 2×10^{-6} and 0.98 for 1×10^{-6} . Second, despite a noticeable downward spike in training loss for all configurations around 60 steps, the validation loss remains smooth and continues to decrease. Third, none of the configurations show signs of overfitting within this training window, as validation loss continues to decrease alongside training loss. We initially trained for 4 epochs but observed that validation loss began to rise after the second epoch, indicating the onset of overfitting. Given the relatively small size of our current dataset, we opted to limit training to 2 epochs for these initial tests. For now, we’ve selected a learning rate of 1×10^{-5} going forward due to its strong validation performance. Additionally, this is the learning rate used in the StreamingVLM paper.



Figure 4.13: Training loss curves for four learning rate configurations over 122 training steps. The higher learning rates (1×10^{-5} and 2×10^{-5}) achieve lower training loss more quickly, while the lower learning rates (1×10^{-6} and 2×10^{-6}) have higher loss. The spike around step 60 suggests potential instability, though the loss recovers in subsequent steps.

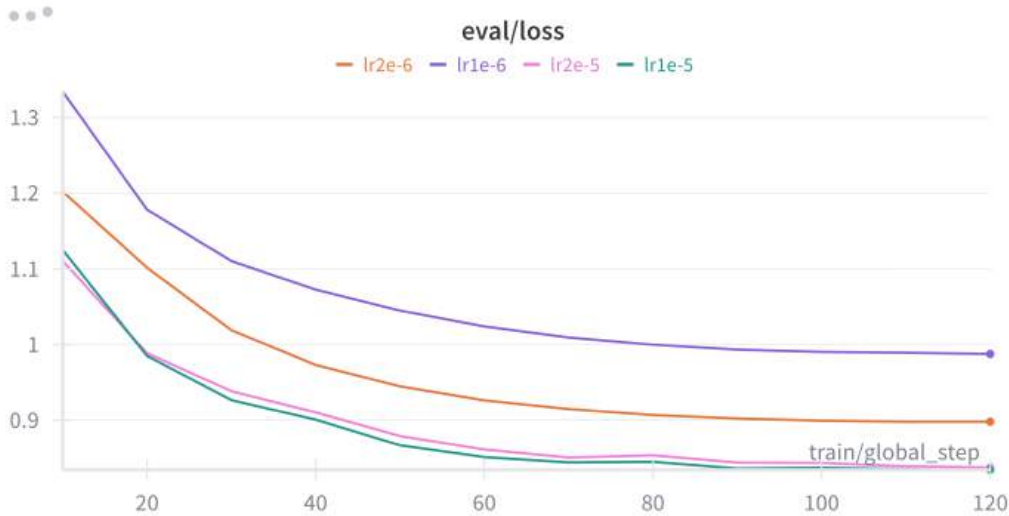


Figure 4.14: Validation loss curves for four learning rate configurations. The higher learning rates (1×10^{-5} and 2×10^{-5}) converge to lower validation loss (approximately 0.85) compared to the lower learning rates, which plateau around 0.89/0.98. All configurations show consistent improvement without signs of overfitting within the 122-step training window.

4.3.2 Qualitative Results

To evaluate the quality of the generated commentary, we run inference with the fine-tuned model. Figure 4.15 shows an example from a Rogers vs. Rybakina match, where the

model correctly identifies both players and generates contextually appropriate play-by-play commentary synchronized with the video.



Figure 4.15: Initial results from fine-tuning StreamingVLM on our tennis dataset. The model generates real-time commentary (shown at bottom) synchronized with the video feed, correctly identifying player names (Rogers, Rybakina) and describing the current action (“Rogers responds with a forehand off the bounce”). This shows that even in small-scale experiments, the model has an ability to generate contextually appropriate commentary.

Figures 4.16 and 4.17 present additional examples from Musetti vs. Fritz and Sinner vs. Shelton matches. The commentary exhibits several desirable properties: it correctly identifies players from the scoreboard, accurately describes shot types (crosscourt forehand, ace down the T), maintains proper score awareness, and varies its phrasing based on the match context.

These qualitative results suggest the even with our relatively modest dataset, the fine-tuned model has learned to generate coherent, contextually appropriate tennis commentary. The model successfully identifies players, describes shots with appropriate tennis terminology, and maintains awareness of match context like the current score. These are all critical components of play-by-play commentary.



Figure 4.16: Sample commentary outputs from a Musetti vs. Fritz match, demonstrating the model's ability to track score progression and generate appropriate commentary for different match moments: point conclusions (a, c) and pre-serve context (b).



Figure 4.17: Sample commentary outputs from a Sinner vs. Shelton match. The model demonstrates recognition of specific shot types such as aces (a) and maintains accurate score tracking throughout the match (b).

4.4 Limitations and Future Work

Our initial fine-tuning results demonstrate that StreamingVLM can learn to generate tennis-specific commentary from a limited set of examples. However, current versions of the model still exhibit several limitations that must be addressed before it can produce broadcast-quality commentary.

First, the commentary cadence is not always natural. The model sometimes generates text too quickly during slow moments, or fails to keep pace with rapid exchanges. Second, the model will produce factually incorrect statements, such as attributing shots to the wrong player or announcing incorrect scores. Third, the model struggles to maintain coherent commentary over extended periods, sometimes losing track of the match narrative or repeating similar phrases.

These limitations likely stem from the small size of our current training dataset. For context, the original StreamingVLM was trained on a much larger amount of streaming samples across two stages: 525K streaming samples plus LiveCC’s Live-WhisperX-526K dataset in the first stage, followed by 14K high-quality annealing samples in the second stage [25]. The sports commentary data alone comprised 712 basketball games, 544 soccer games, 402 ice hockey games, 399 baseball games and 392 American football games. In comparison, our tennis dataset of approximately 2,100 examples across 11 matches represents a small fraction of this scale, limiting the model’s exposure to the full range of match situations, commentary styles and temporal patterns.

To address these limitations, we plan to pursue several directions:

- **Scaling the dataset:** We will apply the Method 2 pipeline to additional matches across other tournaments to increase both the volume and diversity of training examples. We will also explore methods to gather a larger based of Method 1 examples for further diversity.
- **Extended fine-tuning experiments:** We will explore longer training schedules with larger datasets and additional hyperparameter tuning to extract the best performance. If we are able to scale Method 1 to produce more examples, we could implement a two-stage fine-tuning approach similar to the original StreamingVLM training: first fine-tuning on the larger base of Method 2 data, then fine-tuning on the more specific but smaller Method 1 data.

4.4.1 Integration with Prosodic TTS

Once we achieve a strong baseline model for commentary generation, the next phase will be to integrate the output with ProsodyLM for text-to-speech synthesis. The integration may be different than the process described in Section 3.4, due to advancements in ProsodyLM and depending on the output from the fine-tuned StreamingVLM. This will complete the unified pipeline from raw video input to expressive audio commentary.

Chapter 5

Conclusion

This thesis presented a system for generating real-time, emotionally expressive AI commentary for tennis matches. We addressed two fundamental challenges in automated sports commentary: generating accurate, contextually appropriate content synchronized with visual events, and delivering that content with natural prosody that matches the emotional intensity of the match.

5.1 Summary of Contributions

Our first contribution is a modular pipeline for excitement-driven prosodic commentary. This pipeline processes isolated tennis match clips through a computer vision module to extract shot classifications and directions, combines this information with tournament metadata, and uses a large language model to generate timestamped commentary. A post-processing stage adds precise timestamps aligned to shot boundaries and assigns sentence-level excitement scores on a 1–5 scale. The pipeline maps tournament-provided excitement metadata to three discrete levels that control both the linguistic style of the generated text and the prosodic characteristics of the synthesized speech via ProsodyLM. This system was demonstrated at the US Open 2025 Behind the Scenes session, showcasing the shift from monotonic AI commentary to expressive, human-like delivery.

Our second contribution is a dataset and methodology for adapting StreamingVLM to the tennis domain. We developed two complementary data generation methods. The first leverages our modular pipeline to produce high-quality, precisely aligned commentary from isolated point clips, yielding 148 examples with controlled excitement levels. The second implements a scalable four-stage pipeline that extracts commentary from YouTube tennis streams: OCR-based score detection segments matches at point boundaries, ASR transcribes the commentary audio, an alignment stage synchronizes the two streams, and LLM-based

cleaning restructures the raw transcripts into the target format. This approach produced over 2,000 training examples from 11 Wimbledon 2024 matches.

Our third contribution is an initial demonstration of fine-tuning StreamingVLM for tennis commentary. We conducted learning rate experiments across four configurations, finding that higher learning rates (1×10^{-5} and 2×10^{-6}) achieved better validation performance within our small 2-epoch training window. Qualitative evaluation shows that the fine-tuned model can correctly identify players, describe shots with appropriate tennis terminology, and maintain score awareness. These are all critical components of effective play-by-play commentary. The current limitations of the model and directions for future work, including dataset scaling, two-stage fine-tuning, and integration with prosodic TTS, are discussed in Section 4.4.

5.2 Closing Remarks

Automated sports commentary sits at the intersection of computer vision, natural language generation and speech synthesis, which have all seen rapid advancement in recent years. This thesis demonstrates that by combining these technologies thoughtfully, we can begin to bridge the gap between what AI systems can perceive and describe and how human commentators engage audiences through expressive delivery.

While significant work remains before AI commentary can match the quality of experienced human broadcasters, the progress shown here suggests a path towards expanding coverage rather than replacing it. As tournaments continue to expand and fan demand for real-time content grows, AI-generated commentary offers a promising solution for bringing play-by-play coverage to matches that would otherwise go unrated, making the sport more accessible to fans worldwide.

References

- [1] IBM. *IBM and the USTA Roll Out AI-Powered Fan Experiences for 2025 US Open*. IBM Newsroom. Aug. 2025. URL: <https://www.multivu.com/ibm/92400510-en-ibm-usta-ai-fan-experiences-2025-us-open>.
- [2] IBM. *The All England Lawn Tennis Club and IBM Launch New AI Features for Real-Time Wimbledon Fan Engagement*. IBM Newsroom. June 2025. URL: <https://newsroom.ibm.com/2025-06-17-the-all-england-lawn-tennis-club-and-ibm-launch-new-ai-features-for-real-time-wimbledon-fan-engagement>.
- [3] IBM. *IBM and Agassi Sports Entertainment Announce AI-Powered Platform to Advance Global Racquet Sports*. IBM Newsroom. Nov. 2025. URL: <https://newsroom.ibm.com/2025-11-05-ibm-and-agassi-sports-entertainment-announce-ai-powered-platform-to-advance-global-racquet-sports>.
- [4] Morning Consult. *2025 Sports Survey*. Commissioned by IBM. Aug. 2025. URL: https://filecache.mediaroom.com/mr5mr_ibmnewsroom/199459/IBM_Sport_Survey_Report_2025.pdf.
- [5] S. Purohit. “AI Commentator: Narrating Sports Games through Multimodal Perception and Large Language Models.” Master of Engineering Thesis. Massachusetts Institute of Technology, 2023.
- [6] C. Chen, J. Xue, W. Gou, M. Xie, and X. Yao. “Quantitative analysis and evaluation of research on the application of computer vision in sports since the 21st century.” *Frontiers in Sports and Active Living*, **7**, 2025, p. 1604232. DOI: [10.3389/fspor.2025.1604232](https://doi.org/10.3389/fspor.2025.1604232).
- [7] Mordor Intelligence. *Artificial Intelligence Market In Sports – Growth, Trends, and Forecast (2025-2030)*. 2025. URL: <https://www.mordorintelligence.com/industry-reports/artificial-intelligence-market-in-sports>.
- [8] Y. Wu, D. Deng, X. Xie, M. He, J. Xu, H. Zhang, H. Zhang, and Y. Wu. “OBTracker: Visual Analytics of Off-ball Movements in Basketball.” *IEEE Transactions on Visualization and Computer Graphics*, **29**(1), 2023, pp. 929–939. DOI: [10.1109/TVCG.2022.3209373](https://doi.org/10.1109/TVCG.2022.3209373).

- [9] J. Mizels, B. Erickson, and P. Chalmers. “Current State of Data and Analytics Research in Baseball.” *Current Reviews in Musculoskeletal Medicine*, **15**, 2022, pp. 283–290. DOI: [10.1007/s12178-022-09763-6](https://doi.org/10.1007/s12178-022-09763-6). URL: <https://doi.org/10.1007/s12178-022-09763-6>.
- [10] H. Janetzko, D. Sacha, M. Stein, T. Schreck, D. A. Keim, and O. Deussen. “Feature-driven visual analytics of soccer data.” In: *2014 IEEE Conference on Visual Analytics Science and Technology (VAST)*. 2014, pp. 13–22. DOI: [10.1109/VAST.2014.7042477](https://doi.org/10.1109/VAST.2014.7042477).
- [11] T. Polk, D. Jäckle, J. Häußler, and J. Yang. “CourtTime: Generating Actionable Insights into Tennis Matches Using Visual Analytics.” *IEEE Transactions on Visualization and Computer Graphics*, **26**(1), 2020, pp. 397–406. DOI: [10.1109/TVCG.2019.2934243](https://doi.org/10.1109/TVCG.2019.2934243).
- [12] J.-P. Choe, I.-W. Hwang, J.-H. Park, C. Amo, and J.-M. Lee. “How valid is the commercially available tennis match analysis mobile application? Is it good enough?” *International Journal of Performance Analysis in Sport*, Oct. 2023.
- [13] A. Harris. *Artificial intelligence in tennis: As Wimbledon embraces AI, how can players use the technology to improve their game?* Oct. 2024. URL: <https://www.skysports.com/tennis/news/12110/13234287/artificial-intelligence-in-tennis-as-wimbledon-embraces-ai-how-can-players-use-the-technology-to-improve-their-game>.
- [14] J. O’Halloran. *Serving up tennis improvements through AI*. Sept. 2025. URL: <https://www.computerweekly.com/feature/Serving-up-tennis-improvements-through-AI>.
- [15] L. Mercedes. “With electronic line calling and generative AI, Wimbledon’s technology is on point.” *Ingenia*, 2025. URL: <https://www.ingenia.org.uk/articles/wimbledons-technology-is-on-point/>.
- [16] Boulton Wade Tennant. *From umpires to analytics – AI innovation in tennis*. Aug. 2025. URL: <https://www.boulton.com/bulletin/from-umpires-to-analytics-ai-innovation-in-tennis/>.
- [17] NewKerala. *Indian AI to Power Kyrgios-Sabalenka Tennis Match Analytics*. 2025. URL: <https://www.newkerala.com/news/o/indian-startup-kproai-deploy-performance-analytics-kyrgios-sabalenka-battle-116>.
- [18] P. Andrews, N. Borch, and M. Fjeld. “FootyVision: Multi-Object Tracking, Localisation, and Augmentation of Players and Ball in Football Video.” In: *Proceedings of the 2024 9th International Conference on Multimedia and Image Processing. ICMIP ’24*. Osaka, Japan: Association for Computing Machinery, 2024, pp. 15–25. ISBN: 9798400716164. DOI: [10.1145/3665026.3665029](https://doi.org/10.1145/3665026.3665029). URL: <https://doi.org/10.1145/3665026.3665029>.

- [19] T. Kumano, M. Ichiki, K. Kurihara, H. Kaneko, T. Komori, T. Shimizu, N. Seiyama, A. Imai, H. Sumiyoshi, and T. Takagi. “Generation of Automated Sports Commentary from Live Sports Data.” In: *2019 IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB)*. 2019, pp. 1–4. DOI: [10.1109/BMSB47279.2019.8971879](https://doi.org/10.1109/BMSB47279.2019.8971879).
- [20] A. Cook and O. Karakuş. “LLM-Commentator: Novel fine-tuning strategies of large language models for automatic commentary generation using football event data.” *Knowledge-Based Systems*, **300**, 2024, p. 112219. ISSN: 0950-7051. DOI: <https://doi.org/10.1016/j.knosys.2024.112219>. URL: <https://www.sciencedirect.com/science/article/pii/S0950705124008530>.
- [21] J. Rao, H. Wu, C. Liu, Y. Wang, and W. Xie. *MatchTime: Towards Automatic Soccer Game Commentary Generation*. 2024. arXiv: [2406.18530](https://arxiv.org/abs/2406.18530) [cs.CV]. URL: <https://arxiv.org/abs/2406.18530>.
- [22] Z. Li, X. Wu, H. Du, F. Liu, H. Nghiem, and G. Shi. “A Survey of State of the Art Large Vision Language Models: Benchmark Evaluations and Challenges.” In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*. June 2025, pp. 1587–1606.
- [23] X. Li, Y. He, S. Zu, Z. Li, T. Shi, Y. Xie, and K. Zhang. “Multi-Modal Large Language Model with RAG Strategies in Soccer Commentary Generation.” In: *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 2025, pp. 6197–6206. DOI: [10.1109/WACV61041.2025.00604](https://doi.org/10.1109/WACV61041.2025.00604).
- [24] L. You, W. Huang, X. Xie, X. Wei, B. Li, S. Lin, Y. Li, and C. Wang. “TimeSoccer: An End-to-End Multimodal Large Language Model for Soccer Commentary Generation.” *arXiv preprint arXiv:2504.17365*, 2025.
- [25] R. Xu, G. Xiao, Y. Chen, L. He, K. Peng, Y. Lu, and S. Han. *StreamingVLM: Real-Time Understanding for Infinite Video Streams*. 2025. arXiv: [2510.09608](https://arxiv.org/abs/2510.09608) [cs.CV]. URL: <https://arxiv.org/abs/2510.09608>.
- [26] S. Brognaux, B. Picart, and T. Drugman. “A new prosody annotation protocol for live sports commentaries.” In: *Interspeech 2013*. 2013, pp. 1554–1558. DOI: [10.21437/Interspeech.2013-393](https://doi.org/10.21437/Interspeech.2013-393).
- [27] Y. Wang et al. “Tacotron: A Fully End-to-End Text-To-Speech Synthesis Model.” *CoRR*, **abs/1703.10135**, 2017. arXiv: [1703.10135](https://arxiv.org/abs/1703.10135). URL: <http://arxiv.org/abs/1703.10135>.

- [28] Y. Ren, Y. Ruan, X. Tan, T. Qin, S. Zhao, Z. Zhao, and T.-Y. Liu. “FastSpeech: Fast, Robust and Controllable Text to Speech.” In: *Advances in Neural Information Processing Systems*. Ed. by H. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché-Buc, E. Fox, and R. Garnett. Vol. 32. Curran Associates, Inc., 2019. URL: https://proceedings.neurips.cc/paper_files/paper/2019/file/f63f65b503e22cb970527f23c9ad7db1-Paper.pdf.
- [29] Y. Wang, D. Stanton, Y. Zhang, R. J. Skerry-Ryan, E. Battenberg, J. Shor, Y. Xiao, F. Ren, Y. Jia, and R. A. Saurous. “Style Tokens: Unsupervised Style Modeling, Control and Transfer in End-to-End Speech Synthesis.” *CoRR*, **abs/1803.09017**, 2018. arXiv: [1803.09017](http://arxiv.org/abs/1803.09017). URL: <http://arxiv.org/abs/1803.09017>.
- [30] P.-f. Wu, Z.-h. Ling, L.-j. Liu, Y. Jiang, H.-c. Wu, and L.-r. Dai. *End-to-End Emotional Speech Synthesis Using Style Tokens and Semi-Supervised Training*. 2019. arXiv: [1906.10859](https://arxiv.org/abs/1906.10859) [**eess.AS**]. URL: <https://arxiv.org/abs/1906.10859>.
- [31] R. Skerry-Ryan, E. Battenberg, Y. Xiao, Y. Wang, D. Stanton, J. Shor, R. J. Weiss, R. Clark, and R. A. Saurous. *Towards End-to-End Prosody Transfer for Expressive Speech Synthesis with Tacotron*. 2018. arXiv: [1803.09047](https://arxiv.org/abs/1803.09047) [**cs.CL**]. URL: <https://arxiv.org/abs/1803.09047>.
- [32] H. Wang, S. Shan, Y. Guo, and Y. Wang. “ProsodyFlow: High-fidelity Text-to-Speech through Conditional Flow Matching and Prosody Modeling with Large Speech Language Models.” In: *Proceedings of the 31st International Conference on Computational Linguistics*. Abu Dhabi, UAE: Association for Computational Linguistics, Jan. 2025, pp. 7748–7753. URL: <https://aclanthology.org/2025.coling-main.518/>.
- [33] K. Qian, X. Fan, J. Ni, S. Shechtman, M. Hasegawa-Johnson, C. Gan, and Y. Zhang. “ProsodyLM: Uncovering the Emerging Prosody Processing Capabilities in Speech Language Models.” In: *Proceedings of the Conference on Language Modeling (COLM)*. 2025. URL: <https://arxiv.org/abs/2507.20091>.
- [34] A. McConnon. *The future of tennis broadcasting: Excitement-driven AI sports commentary*. Nov. 2025. URL: <https://www.ibm.com/think/news/future-tennis-broadcasting-ai-sports-commentary>.
- [35] NVIDIA. *Parakeet TDT: Transducer-based Speech Recognition Model*. Hugging Face Model Hub. 2023. URL: <https://huggingface.co/nvidia/parakeet-tdt-0.6b-v2>.